However, under the conditions of complete electron degeneracy, there are no vacant states available for an emitted electron to occupy, so the neutrons cannot decay back into protons.[24]

When the density reaches about 4×10^{14} kg m^{-3}, the minimum-energy arrangement is one in which some of the neutrons are found *outside* the nuclei. The appearance of these free neutrons is called **neutron drip** and marks the start of a three-component mixture of a lattice of neutron-rich nuclei, nonrelativistic degenerate free neutrons, and relativistic degenerate electrons.

The fluid of free neutrons has the striking property that it has no viscosity. This occurs because a spontaneous pairing of the degenerate neutrons has taken place. The resulting combination of two fermions (the neutrons) is a boson (recall Section 5.4) and so is not subject to the restrictions of the Pauli exclusion principle. Because degenerate bosons can *all* crowd into the lowest energy state, the fluid of paired neutrons can lose no energy. It is a **superfluid** that flows without resistance. Any whirlpools or vortices in the fluid will continue to spin forever without stopping.

As the density increases further, the number of free neutrons increases as the number of electrons declines. The neutron degeneracy pressure exceeds the electron degeneracy pressure when the density reaches roughly 4×10^{15} kg m^{-3}. As the density approaches $\rho_{\rm nuc}$, the nuclei effectively dissolve as the distinction between neutrons inside and outside of nuclei becomes meaningless. This results in a fluid mixture of free neutrons, protons, and electrons dominated by neutron degeneracy pressure, with both the neutrons and protons paired to form superfluids. The fluid of pairs of positively charged protons is also **superconducting**, with zero electrical resistance. As the density increases further, the ratio of neutrons:protons:electrons approaches a limiting value of 8:1:1, as determined by the balance between the competing processes of electron capture and β-decay inhibited by the presence of degenerate electrons.

The properties of the neutron star material when $\rho > \rho_{\rm nuc}$ are still poorly understood. A complete theoretical description of the behavior of a sea of free neutrons interacting via the strong nuclear force in the presence of protons and electrons is not yet available, and there is little experimental data on the behavior of matter in this density range. A further complication is the appearance of sub-nuclear particles such as *pions* (π) produced by the decay of a neutron into a proton and a negatively charged pion, $n \rightarrow p^+ + \pi^-$, which occurs spontaneously in neutron stars when $\rho > 2\rho_{\rm nuc}$.[25] Nevertheless, these are the values of the density encountered in the interiors of neutron stars, and the difficulties mentioned are the primary reasons for the uncertainty in the structure calculated for model neutron stars.

Neutron Star Models

Table 16.1 summarizes the composition of the neutron star material at various densities. After an equation of state that relates the density and pressure has been obtained, a model of the star can be calculated by numerically integrating general-relativistic versions of the

[24]An *isolated* neutron decays into a proton in about 10.2 minutes, the half-life for that process.

[25]The π^- is a negatively charged particle that is 273 times more massive than the electron. It mediates the strong nuclear force that holds an atomic nucleus together. (The strong force between nucleons was described in Section 10.3.) Pions have been produced and studied in high-energy accelerator laboratories.

TABLE 16.1 Composition of Neutron Star Material.

Transition density $(\mathrm{kg\ m^{-3}})$	Composition	Degeneracy pressure
	iron nuclei, nonrelativistic free electrons	electron
$\approx 1 \times 10^{9}$	electrons become relativistic	
	iron nuclei, relativistic free electrons	electron
$\approx 1 \times 10^{12}$	neutronization	
	neutron-rich nuclei, relativistic free electrons	electron
$\approx 4 \times 10^{14}$	neutron drip	
	neutron-rich nuclei, free neutrons, relativistic free electrons	electron
$\approx 4 \times 10^{15}$	neutron degeneracy pressure dominates	
	neutron-rich nuclei, superfluid free neutrons, relativistic free electrons	neutron
$\approx 2 \times 10^{17}$	nuclei dissolve	
	superfluid free neutrons, superconducting free protons, relativistic free electrons	neutron
$\approx 4 \times 10^{17}$	pion production	
	superfluid free neutrons, superconducting free protons, relativistic free electrons, other elementary particles (pions, …?)	neutron

stellar structure equations collected at the beginning of Section 10.5. The first quantitative model of a neutron star was calculated by J. Robert Oppenheimer (1904–1967) and G. M. Volkoff (1914–2000) at Berkeley in 1939. Figure 16.11 shows the result of a recent calculation of a 1.4 $M_\odot$ neutron star model. Although the details are sensitive to the equation of state used, this model displays some typical features.

1. The outer crust consists of heavy nuclei, in the form of either a fluid "ocean" or a solid lattice, and relativistic degenerate electrons. Nearest the surface, the nuclei are probably $^{56}_{26}\mathrm{Fe}$. At greater depth and density, increasingly neutron-rich nuclei are encountered until neutron drip begins at the bottom of the outer crust (where $\rho \approx 4 \times 10^{14}\ \mathrm{kg\ m^{-3}}$).
2. The inner crust consists of a three-part mixture of a lattice of nuclei such as $^{118}_{36}\mathrm{Kr}$, a superfluid of free neutrons, and relativistic degenerate electrons. The bottom of the inner crust occurs where $\rho \approx \rho_{\mathrm{nuc}}$, and the nuclei dissolve.

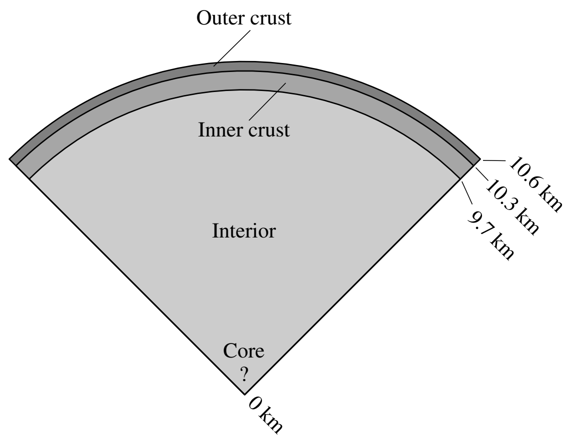

FIGURE 16.11 A 1.4 $M_\odot$ neutron star model.

3. The interior of the neutron star consists primarily of superfluid neutrons, with a smaller number of superfluid, superconducting protons and relativistic degenerate electrons.
4. There may or may not be a solid core consisting of pions or other sub-nuclear particles. The density at the center of a 1.4 $M_\odot$ neutron star is about 10^{18} kg m^{-3}.

The Chandrasekhar Limit for Neutron Stars

Like white dwarfs, neutron stars obey a mass–volume relation,

$$\boxed{M_{\rm ns} V_{\rm ns} = \text{constant},} \tag{16.25}$$

so neutron stars become smaller and more dense with increasing mass. However, this mass–volume relation fails for more massive neutron stars because there is a point beyond which neutron degeneracy pressure can no longer support the star. Hence, there is a maximum mass for neutron stars, analogous to the Chandrasekhar mass for white dwarfs. As might be expected, the value of this maximum mass is different for different choices of the equation of state. However, detailed computer modeling of neutron stars, along with a very general argument involving the general theory of relativity, shows that the maximum mass possible for a neutron star cannot exceed about 2.2 $M_\odot$ if it is static, and 2.9 $M_\odot$ if it is rotating rapidly.[26] If a neutron star is to remain dynamically stable and resist collapsing, it must be able to respond to a small disturbance in its structure by rapidly adjusting its pressure to compensate. However, there is a limit to how quickly such an adjustment can be made because these changes are conveyed by sound waves that must move more slowly than light. If a neutron star's mass exceeds 2.2 $M_\odot$ in the static case or 2.9 $M_\odot$ in the rapidly rotating case, it cannot generate pressure quickly enough to avoid collapsing. The result is a black hole (as will be discussed in Section 17.3).

[26]Recall from Section 15.4 that centrifugal effects provide additional support to a rapidly rotating neutron star.

Rapid Rotation and Conservation of Angular Momentum

Several properties of neutron stars were anticipated before they were observed. For example, neutron stars must rotate very rapidly. If the iron core of the pre-supernova supergiant star were rotating even slowly, the decrease in radius would be so great that the conservation of angular momentum would guarantee the formation of a rapidly rotating neutron star.

The scale of the collapse can be found from Eqs. (16.13) and (16.24) for the estimated radii of a white dwarf and neutron star if we assume that the progenitor core is characteristic of a white dwarf composed entirely of iron. Although the leading constants in both expressions are spurious (a by-product of the approximations made), the *ratio* of the radii is more accurate:

$$\frac{R_{\rm core}}{R_{\rm ns}} \approx \frac{m_n}{m_e}\left(\frac{Z}{A}\right)^{5/3} = 512,$$

where $Z/A = 26/56$ for iron has been used. Now apply the conservation of angular momentum to the collapsing core (which is assumed here for simplicity to lose no mass, so $M_{\rm core} = M_{\rm wd} = M_{\rm ns}$). Treating each star as a sphere with a moment of inertia of the form $I = CMR^2$, we have[27]

$$\begin{aligned} I_i\omega_i &= I_f\omega_f \\ CM_iR_i^2\omega_i &= CM_fR_f^2\omega_f \\ \omega_f &= \omega_i\left(\frac{R_i}{R_f}\right)^2. \end{aligned}$$

In terms of the rotation period P, this is

$$P_f = P_i\left(\frac{R_f}{R_i}\right)^2. \tag{16.26}$$

For the specific case of an iron core collapsing to form a neutron star, Eq. (16.6) shows that

$$P_{\rm ns} \approx 3.8 \times 10^{-6}\ P_{\rm core}. \tag{16.27}$$

The question of how fast the progenitor core may be rotating is difficult to answer. As a star evolves, its contracting core is not completely isolated from the surrounding envelope, so one cannot use the simple approach to conservation of angular momentum described above.[28] For purposes of estimation, we will take $P_{\rm core} = 1350$ s, the rotation period observed for the white dwarf 40 Eridani B (shown in the H–R diagrams of Figs. 8.12 and 8.16). Inserting this into Eq. (16.27) results in a rotation period of about 5×10^{-3} s. Thus neutron stars will be rotating very rapidly when they are formed, with rotation periods on the order of a few milliseconds.

[27]The constant C is determined by the distribution of mass inside the star. For example, $C = 2/5$ for a uniform sphere. We assume that the progenitor core and neutron star have about the same value of C.

[28]The core and envelope may exchange angular momentum by magnetic fields or rotational mixing via the very slow *meridional currents* that generally circulate upward at the poles and downward at the equator of a rotating star.

"Freezing In" Magnetic Field Lines

Another property predicted for neutron stars is that they should have extremely strong magnetic fields. The "freezing in" of magnetic field lines in a conducting fluid or gas (mentioned in Section 11.3 in connection with sunspots) implies that the *magnetic flux* through the surface of a white dwarf will be conserved as it collapses to form a neutron star. The flux of a magnetic field through a surface $\mathcal{S}$ is defined as the surface integral

$$\Phi \equiv \int_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{A},$$

where $\mathbf{B}$ is the magnetic field vector (see Fig. 16.12). In approximate terms, if we ignore the geometry of the magnetic field, this means that the product of the magnetic field strength and the area of the star's surface remains constant. Thus

$$B_i 4\pi R_i^2 = B_f 4\pi R_f^2. \tag{16.28}$$

In order to use Eq. (16.28) to estimate the magnetic field of a neutron star, we must first know what the strength of the magnetic field is for the iron core of a pre-supernova star. Although this is not at all clear, we can use the largest observed white-dwarf magnetic field of $B \approx 5 \times 10^4$ T as an extreme case, which is large compared to a typical white-dwarf magnetic field of perhaps 10 T, and huge compared with the Sun's global field of about 2×10^{-4} T. Then, using Eq. (16.6), the magnetic field of the neutron star would be

$$B_{\text{ns}} \approx B_{\text{wd}} \left(\frac{R_{\text{wd}}}{R_{\text{ns}}} \right)^2 = 1.3 \times 10^{10} \text{ T}.$$

This shows that neutron stars could be formed with extremely strong magnetic fields, although smaller values such as 10^8 T or less are more typical.

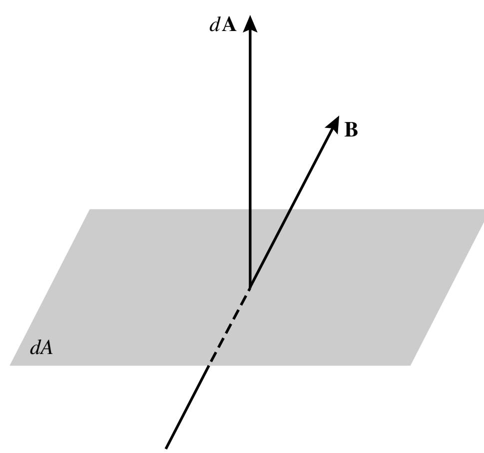

FIGURE 16.12 Magnetic flux, $d\Phi = \mathbf{B} \cdot d\mathbf{A}$, through an element of surface area $d\mathbf{A}$.

Neutron Star Temperatures

The final property of neutron stars is the most obvious. They were extremely hot when they were forged in the "fires" of a supernova, with $T \sim 10^{11}$ K. During the first day, the neutron star cools by emitting neutrinos via the so-called **URCA process**,[29]

$$n \to p^+ + e^- + \overline{\nu}_e$$

$$p^+ + e^- \to n + \nu_e.$$

As the nucleons shuttle between being neutrons and being protons, large numbers of neutrinos and antineutrinos are produced that fly unhindered into space, carrying away energy and thus cooling the neutron star. This process can continue only as long as the nucleons are not degenerate, and it is suppressed after the protons and neutrons settle into the lowest unoccupied energy states. This degeneracy occurs about one day after the formation of the neutron star, when its internal temperature has dropped to about 10^9 K. Other neutrino-emitting processes continue to dominate the cooling for approximately the first thousand years, after which photons emitted from the star's surface take over. The neutron star is a few hundred years old when its internal temperature has declined to 10^8 K, with a surface temperature of several million K. By now the cooling has slowed considerably, and the surface temperature will hover around 10^6 K for the next ten thousand years or so as the neutron star cools at an essentially constant radius.

It is interesting to calculate the blackbody luminosity of a 1.4 $M_\odot$ neutron star with a surface temperature of $T = 10^6$ K. From the Stefan–Boltzmann law, Eq. (3.17),

$$L = 4\pi R^2 \sigma T_e^4 = 7.13 \times 10^{25} \text{ W}.$$

Although this is comparable to the luminosity of the Sun, the radiation is primarily in the form of X-rays since, according to Wien's displacement law, Eq. (3.19),

$$\lambda_{\max} = \frac{(500 \text{ nm})(5800 \text{ K})}{T} = 2.9 \text{ nm}.$$

Prior to the advent of X-ray observatories such as ROSAT, ASCA, and Chandra, astronomers held little hope of ever observing such an exotic object, barely the size of San Diego, California.

16.7 ■ PULSARS

Jocelyn Bell spent two years setting up a forest of 2048 radio dipole antennae over four and a half acres of English countryside. She and her Ph.D. thesis advisor, Anthony Hewish, were using this radio telescope, tuned to a frequency of 81.5 MHz, to study the scintillation ("flickering") that is observed when the radio waves from distant sources known as quasars

[29]The URCA process, which efficiently removes energy from a hot neutron star, is named for the Casino de URCA in Rio de Janeiro, in remembrance of the efficiency with which it removed money from an unlucky physicist. The casino was closed by Brazil in 1955.

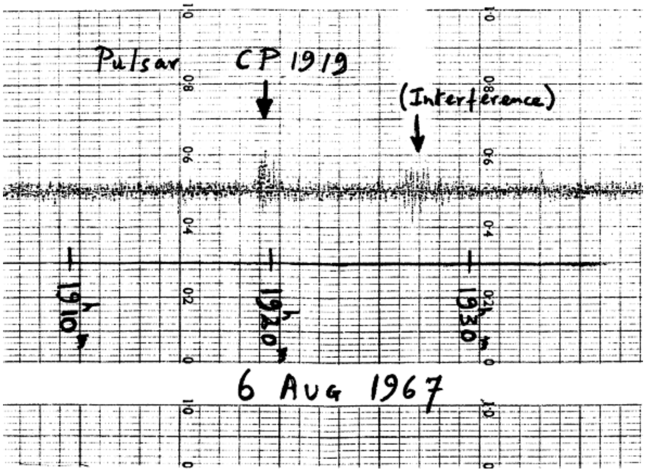

FIGURE 16.13 Discovery of the first pulsar, PSR 1919+21 ("CP" stands for Cambridge Pulsar). (Figure from Lyne and Graham-Smith, *Pulsar Astronomy*, ©Cambridge University Press, New York, 1990. Reprinted with the permission of Cambridge University Press.)

pass through the solar wind. In July 1967, Bell was puzzled to find a bit of "scruff" that reappeared every 400 feet or so on the rolls of her strip chart recorder; see Fig. 16.13. Careful measurements showed that this quarter inch of ink reappeared every 23 hours and 56 minutes, indicating that its source passed over her fixed array of antennae once every sidereal day. Bell concluded that the source was out among the stars rather than within the Solar System. To better resolve the signal, she used a faster recorder and discovered that the scruff consisted of a series of regularly spaced radio pulses 1.337 s apart (the pulse **period**, P). Such a precise celestial clock was unheard of, and Bell and Hewish considered the possibility that these might be signals from an extraterrestrial civilization. If this were true, she felt annoyed that the aliens had chosen such an inconvenient time to make contact. She recalled, "I was now two and a half years through a three year studentship and here was some silly lot of Little Green Men using *my* telescope and *my* frequency to signal to planet Earth." When Bell found another bit of scruff, coming from another part of the sky, her relief was palpable. She wrote, "It was highly unlikely that two lots of Little Green Men could choose the same unusual frequency and unlikely technique to signal to the same inconspicuous planet Earth!"

Hewish, Bell, and their colleagues announced the discovery of these mysterious **pulsars**,[30] and several more were quickly found by other radio observatories. At the time this text was written, more than 1500 pulsars were known, and each is designated by a "PSR"

[30]The term *pulsar* was coined by the science correspondent for the London *Daily Telegraph*. See Hewish et al. (1968) for details of the discovery of pulsars. In 1974 Hewish was awarded a share of the Nobel Prize, along with Martin Ryle (1918–1984), for their work in radio astronomy. Fred Hoyle (1915–2001) and others have argued that Jocelyn Bell should have shared the prize as well; Hewish had designed the radio array and observational technique, but Bell was the first to notice the pulsar signal. This controversial omission has inspired references to the award as the "no-Bell" prize.

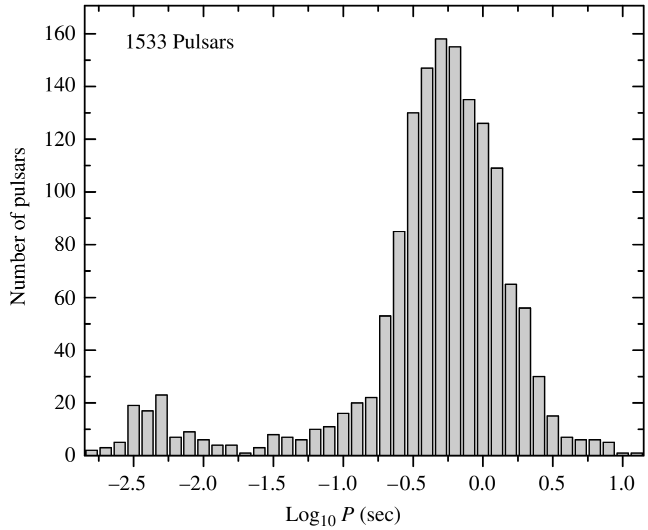

FIGURE 16.14 The distribution of periods for 1533 pulsars. The millisecond pulsars are clearly evident on the left. The average period is about 0.795 s. (Data from Manchester, R. N., Hobbs, G. B., Teoh, A., and Hobbs, M., *A. J.*, *129*, 1993, 2005. Data available at `http://www.atnf.csiro.au/research/pulsar/psrcat`.)

prefix (for *P*ulsating *S*ource of *R*adio) followed by its right ascension (α) and declination (δ). For example, the source of Bell's scruff is PSR 1919+21, identifying its position as $\alpha = 19^{\mathrm{h}}19^{\mathrm{m}}$ and $\delta = +21^\circ$.

General Characteristics

All known pulsars share the following characteristics, which are crucial clues to their physical nature:

- Most pulsars have periods between 0.25 s and 2 s, with an average time between pulses of about 0.795 s (see Fig. 16.14). The pulsar with the longest known period is PSR 1841-0456 ($P = 11.8$ s); Terzan 5ad (PSR J1748-2446ad) is the fastest known pulsar ($P = 0.00139$ s).

- Pulsars have extremely well-defined pulse periods and would make exceptionally accurate clocks. For example, the period of PSR 1937+214 has been determined to be $P = 0.00155780644887275$ s, a measurement that challenges the accuracy of the best atomic clocks. (Such precise determinations are possible because of the enormous number of pulsar measurements that can be made, given their very short periods.)

- The periods of all pulsars increase very gradually as the pulses slow down, the rate of increase being given by the period derivative $\dot{P} \equiv dP/dt$.[31] Typically, $\dot{P} \approx 10^{-15}$,

[31] Note that $\dot{P}$ is measured in terms of seconds of period change per second and so is unitless.

and the *characteristic lifetime* (the time it would take the pulses to cease if $\dot{P}$ were constant) is $P/\dot{P} \approx$ a few 10^7 years. The value of $\dot{P}$ for PSR 1937+214 is unusually small, $\dot{P} = 1.051054 \times 10^{-19}$. This corresponds to a characteristic lifetime of $P/\dot{P} = 1.48 \times 10^{16}$ s, or about 470 million years.

Possible Pulsar Models

These characteristics enabled astronomers to deduce the basic components of pulsars. In the paper announcing their discovery, Hewish, Bell, and their co-authors suggested that an oscillating neutron star might be involved, but American astronomer Thomas Gold (1920–2004) quickly and convincingly argued instead that pulsars are rapidly rotating neutron stars.

There are three obvious ways of obtaining rapid regular pulses in astronomy:

1. **Binary stars**. If the orbital periods of a binary star system are to fall in the range of the observed pulsar periods, then extremely compact stars must be involved—either white dwarfs or neutron stars. The general form of Kepler's third law, Eq. (2.37), shows that if two 1 $M_\odot$ stars were to orbit each other every 0.79 s (the average pulsar period), then their separation would be only 1.6×10^6 m. This is much less than the 5.5×10^6 m radius of Sirius B, and the separation would be even smaller for more rapid pulsars. This eliminates even the smallest, most massive white dwarfs from consideration.

   Neutron stars are so small that two of them could orbit each other with a period in agreement with those observed for pulsars. However, this possibility is ruled out by Einstein's general theory of relativity. As the two neutron stars rapidly move through space and time, gravitational waves are generated that carry energy away from the binary system. As the neutron stars slowly spiral closer together, their orbital period *decreases*, according to Kepler's third law. This contradicts the observed *increase* in the periods of the pulsars and so eliminates binary neutron stars as a source of the radio pulses.[32]

2. **Pulsating stars**. As we noted in Section 16.2, white dwarfs oscillate with periods between 100 and 1000 s. The periods of these nonradial g-modes are much longer than the observed pulsar periods. Of course, it might be imagined that a radial oscillation is involved with the pulsars. However, the period for the radial fundamental mode is a few seconds, too long to explain the faster pulses.

   A similar argument eliminates neutron star oscillations. Neutron stars are about 10^8 times more dense than white dwarfs. According to the period–mean density relation for stellar pulsation (recall Section 14.2), the period of oscillation is proportional to $1/\sqrt{\rho}$. This implies that neutron stars should vibrate approximately 10^4 times more rapidly than white dwarfs, with a radial fundamental mode period around 10^{-4} s and nonradial g-modes between 10^{-2} s and 10^{-1} s. These periods are much too short for the slower pulsars.

[32]Gravitational waves will be described in more detail in Section 18.6, as will the binary system of two neutron stars in which these waves have been indirectly detected.

3. **Rotating stars**. The enormous angular momentum of a rapidly rotating compact star would guarantee its precise clock-like behavior. But how fast can a star spin? Its angular velocity, ω, is limited by the ability of gravity to supply the centripetal force that keeps the star from flying apart. This constraint is most severe at the star's equator, where the stellar material moves most rapidly. Ignore the inevitable equatorial bulging caused by rotation and assume that the star remains circular with radius R and mass M. Then the maximum angular velocity may be found by equating the centripetal and gravitational accelerations at the equator,

$$\omega_{\rm max}^2 R = G\,\frac{M}{R^2},$$

so that the minimum rotation period is $P_{\rm min} = 2\pi/\omega_{\rm max}$, or

$$P_{\rm min} = 2\pi\sqrt{\frac{R^3}{GM}}. \tag{16.29}$$

For Sirius B, $P_{\rm min} \approx 7$ s, which is much too long. However, for a 1.4 $M_\odot$ neutron star, $P_{\rm min} \approx 5 \times 10^{-4}$ s. Because this is a *minimum* rotation time, it can accommodate the complete range of periods observed for pulsars.

Pulsars as Rapidly Rotating Neutron Stars

Only one alternative has emerged unscathed from this process of elimination, namely, that pulsars are rapidly rotating neutron stars. This conclusion was strengthened by the discovery in 1968 of pulsars associated with the Vela and Crab supernovae remnants. (Today dozens of pulsars are known to be associated with supernova remnants.) In addition, the Crab pulsar PSR 0531-21 has a very short pulse period of only 0.0333 s. No white dwarf could rotate 30 times per second without disintegrating, and the last doubts about the identity of pulsars were laid to rest. Until the discovery of the **millisecond pulsars** ($P \approx 10$ ms or less) in 1982, the Crab pulsar held the title of the fastest known pulsar (see Fig. 16.14).[33] The Vela and Crab pulsars not only produce radio bursts but also pulse in other regions of the electromagnetic spectrum ranging from radio to gamma rays, including visible flashes as shown in Fig. 16.15. These young pulsars (and a few others) also display **glitches** when their periods abruptly *decrease* by a tiny amount ($|\Delta P|/P \approx 10^{-6}$ to 10^{-8}); see Fig. 16.16.[34] These sudden spinups are separated by uneven intervals of several years.

Geminga

The nearest pulsar yet detected is only some 90 pc away. PSR 0633+1746, nicknamed Geminga, was well known as a strong source of gamma rays for 17 years before its identity as a pulsar was established in 1992.[35] With a period of 0.237 s, Geminga pulses in both

[33] It is likely that the millisecond pulsars have rapid rotation periods that are a consequence of their membership in close binary systems; more than half of the known millisecond pulsars belong to binaries. For this reason, millisecond pulsars will be discussed in more detail in Section 18.6.

[34] See page 602 for a discussion of possible glitch mechanisms.

[35] *Geminga* means "does not exist" in Milanese dialect, accurately reflecting its long-mysterious nature.

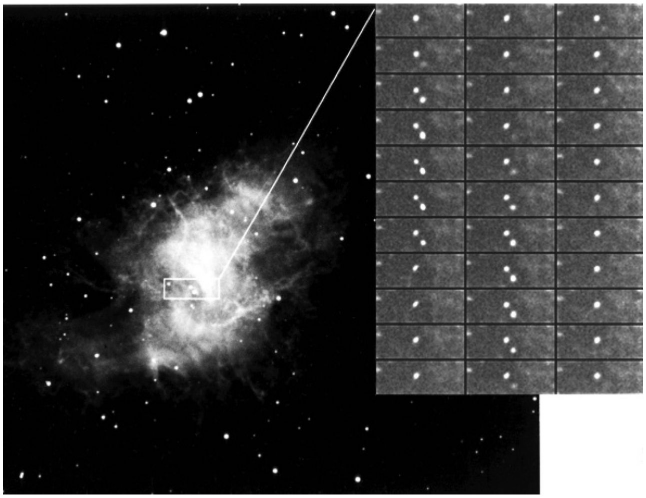

FIGURE 16.15 A sequence of images showing the flashes at visible wavelengths from the Crab pulsar, located at the center of the Crab Nebula (left). A foreground star can be seen as the constant point of light above and to the left of the Crab pulsar. (Courtesy of National Optical Astronomy Observatories.)

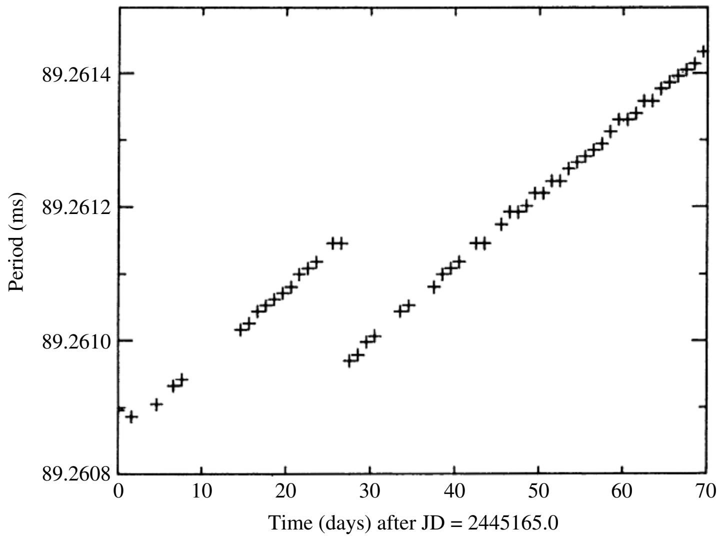

FIGURE 16.16 A glitch in the Vela pulsar. (Figure adapted from McCulloch et al., *Aust. J. Phys.*, *40*, 725, 1987.)

gamma and X-rays (but not at radio wavelengths) and may display glitches. In visible light, its absolute magnitude is fainter than +23.

Evidence for a Core-Collapse Supernova Origin

Although at least one-half of all stars in the sky are known to be members of multiple-star systems, only a few percent of pulsars are known to belong to binary systems. Pulsars also move much faster through space than do normal stars, sometimes with speeds in excess of 1000 km s^{-1}. Both of these observations are consistent with a supernova origin for pulsars. This is because it is highly likely that a core-collapse supernova explosion is not perfectly spherically symmetric, so the forming pulsar could receive a kick, possibly ejecting it from any binary system that it may have been a part of initially. One hypothesis is that the pulsar is formed with an associated asymmetric jet and that, like a jet engine, the pulsar jet could launch the pulsar at high speed away from its formation point.

Synchrotron and Curvature Radiation

Observations of the Crab Nebula, the remnant of the A.D. 1054 supernova, clearly reveal its intimate connection with the pulsar at its center. As shown in Fig. 16.15, the expanding nebula produces a ghostly glow surrounding gaseous filaments that wind throughout it. Interestingly, if the present rate of expansion is extrapolated backward in time, the nebula converges to a point about 90 years *after* the supernova explosion was observed. Obviously the nebula must have been expanding more slowly in the past than it is now, which implies that the expansion is actually accelerating.

In 1953, the Russian astronomer I. Shklovsky (1916–1985) proposed that the white light is **synchrotron radiation** produced when relativistic electrons spiral along magnetic field lines. From the equation for the magnetic force on a moving charge q,

$$\mathbf{F}_m = q(\mathbf{v} \times \mathbf{B}),$$

the component of an electron's velocity $\mathbf{v}$ perpendicular to the field lines produces a circular motion around the lines, while the component of the velocity along the lines is not affected; see Fig. 16.17. As they follow the curved field lines, the relativistic electrons accelerate and emit electromagnetic radiation. It is called synchrotron radiation if the circular motion around the field lines dominates or **curvature radiation** if the motion is primarily along the field lines. In both cases, the shape of the continuous spectrum produced depends on the energy distribution of the emitting electrons and so is easily distinguished from the spectrum of blackbody radiation.[36] The radiation is strongly linearly polarized in the plane of the circular motion for synchrotron radiation and is strongly linearly polarized in the plane of the curving magnetic field line for curvature radiation. As a test of his theory, Shklovsky predicted that the white light from the Crab Nebula would be found to be strongly linearly

[36]Both synchrotron and curvature radiation are sometimes called *nonthermal* to distinguish them from the thermal origin of blackbody radiation.

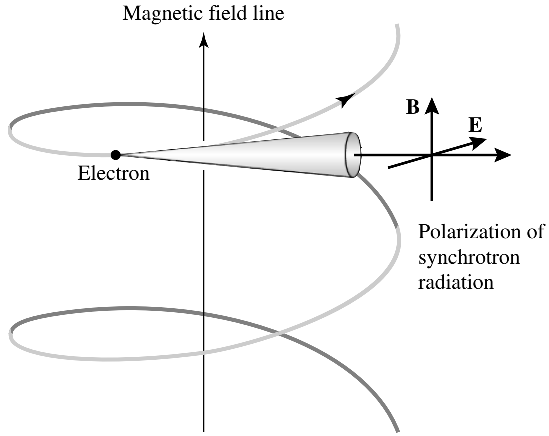

FIGURE 16.17 Synchrotron radiation emitted by a relativistic electron as it spirals around a magnetic field line.

polarized. His prediction was subsequently confirmed as the light from some emitting regions of the nebula was measured to be 60% linearly polarized.

The Energy Source for the Crab's Synchrotron Radiation

The identification of the white glow as synchrotron radiation raised new questions. It implied that magnetic fields of 10^{-7} T must permeate the Crab Nebula. This was puzzling because, according to theoretical estimates, long ago the expansion of the nebula should have weakened the magnetic field far below this value. Furthermore, the electrons should have radiated away all of their energy after only 100 years. It is clear that the production of synchrotron radiation today requires both a replenishment of the magnetic field and a continuous injection of new energetic electrons. The total power needed for the accelerating expansion of the nebula, the relativistic electrons, and the magnetic field is calculated to be about 5×10^{31} W, or more than $10^5\ \mathrm{L}_\odot$.

The energy source is the rotating neutron star at the heart of the Crab Nebula. It acts as a huge flywheel and stores an immense amount of rotational kinetic energy. As the star slows down, its energy supply decreases.

To calculate the rate of energy loss, write the rotational kinetic energy in terms of the period and moment of inertia of the neutron star:

$$K = \frac{1}{2}I\omega^2 = \frac{2\pi^2 I}{P^2}.$$

Then the rate at which the rotating neutron star is losing energy is

$$\frac{dK}{dt} = -\frac{4\pi^2 I \dot{P}}{P^3}. \tag{16.30}$$

Example 16.7.1. Assuming that the neutron star is a uniform sphere with $R = 10$ km and $M = 1.4\ \mathrm{M}_\odot$, its moment of inertia is approximately

$$I = \frac{2}{5} M R^2 = 1.1 \times 10^{38}\ \mathrm{kg\ m^2}.$$

Inserting $P = 0.0333$ s and $\dot{P} = 4.21 \times 10^{-13}$ for the Crab pulsar gives $dK/dt \approx 5.0 \times 10^{31}$ W. Remarkably, this is exactly the energy required to power the Crab Nebula. The slowing down of the neutron star flywheel has enabled the nebula to continue shining and expanding for nearly 1000 years.

It is important to realize that this energy is not transported to the nebula by the pulse itself. The radio luminosity of the Crab's pulse is about 10^{24} W, 200 million times smaller than the rate at which energy is delivered to the nebula. (For older pulsars, the radio pulse luminosity is typically 10^{-5} of the spin-down rate of energy loss.) Thus the pulse process, whatever it may be, is a minor component of the total energy-loss mechanism.

Figure 16.18 shows an HST view of the immediate environment of the Crab pulsar. The ring-like halo seen on the west side of the pulsar is a glowing torus of gas; it may be the result of a polar jet from the pulsar forcing its way through the surrounding nebula. Just to the east of the pulsar, about 1500 AU away, is a bright knot of emission from shocked material in the jet, perhaps due to an instability in the jet itself. Another knot is seen at

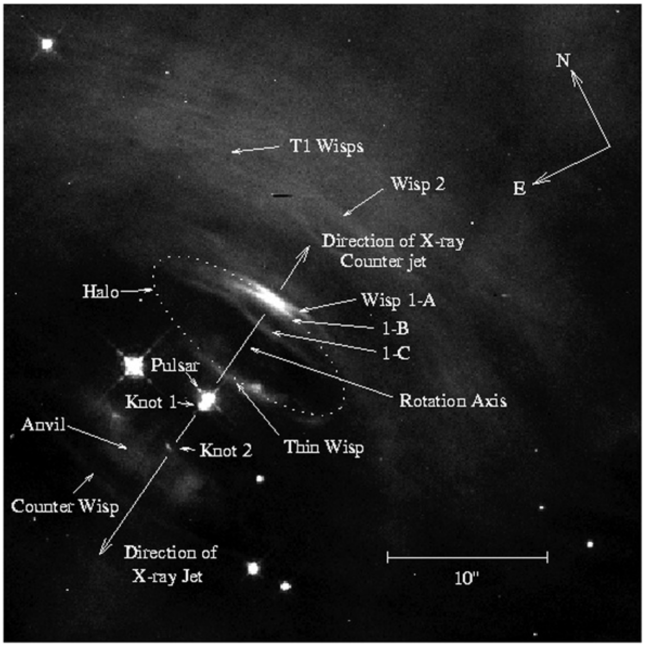

FIGURE 16.18 An HST image of the immediate surroundings of the Crab pulsar. (Figure from Hester et al., *Ap. J.*, *448*, 240, 1995.)

a distance of 9060 AU. Low time-resolution "movies" of the central region of the Crab supernova remnant obtained by long-term observations by HST and Chandra are actually able to show the expansion and evolution of that portion of the nebula. Some of the wisps appear to moving outward at between $0.35c$ and $0.5c$.[37]

The Structure of the Pulses

Before describing the details of a model pulsar, it is worth taking a closer look at the pulses themselves. As can be seen in Fig. 16.19, the pulses are brief and are received over a small fraction of the pulse period (typically from 1% to 5%). Generally, they are received at radio wave frequencies between roughly 20 MHz and 10 GHz.

As the pulses travel through interstellar space, the time-varying electric field of the radio waves causes the electrons that are encountered along the way to vibrate. This process slows the radio waves below the speed of light in a vacuum, c, with a greater retardation at lower frequencies. Thus a sharp pulse emitted at the neutron star, with all frequencies peaking at the same time, is gradually drawn out or *dispersed* as it travels to Earth (see Fig. 16.20). Because more distant pulsars exhibit a greater pulse dispersion, these time delays can be used to measure the distances to pulsars. The results show that the known pulsars are concentrated within the plane of our Milky Way Galaxy (Fig. 16.21) at typical distances of hundreds to thousands of parsecs.

Figure 16.22 shows that there is a substantial variation in the shape of the individual pulses received from a given pulsar. Although a typical pulse consists of a number of brief *subpulses*, the *integrated pulse profile*, an average built up by adding together a train of 100 or more pulses, is remarkably stable. Some pulsars have more than one average pulse profile and abruptly switch back and forth between them (Fig. 16.23). The subpulses may appear at random times in the "window" of the main pulse, or they may march across in a phenomenon known as *drifting subpulses*, as shown in Fig. 16.24. For about 30% of all known pulsars, the individual pulses may simply disappear or *null*, only to reappear up to 100 periods later. Drifting subpulses may even emerge from a nulling event in step with those that entered the null. Finally, the radio waves of many pulsars are strongly linearly polarized (up to 100%), a feature that indicates the presence of a strong magnetic field.

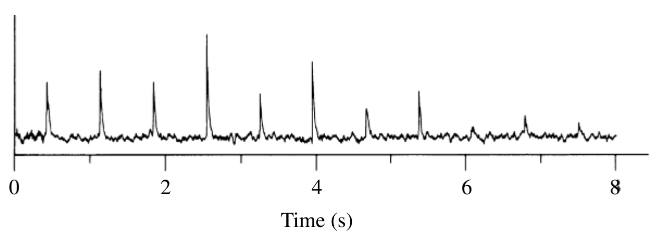

FIGURE 16.19 Pulses from PSR 0329+54 with a period of 0.714 s. (Figure adapted from Manchester and Taylor, *Pulsars*, W. H. Freeman and Co., New York, 1977.)

[37] See Hester, et al., *Ap. J.*, *577*, L49, 2002. The movies are at http://chandra.harvard.edu/photo/2002/0052/movies.html.

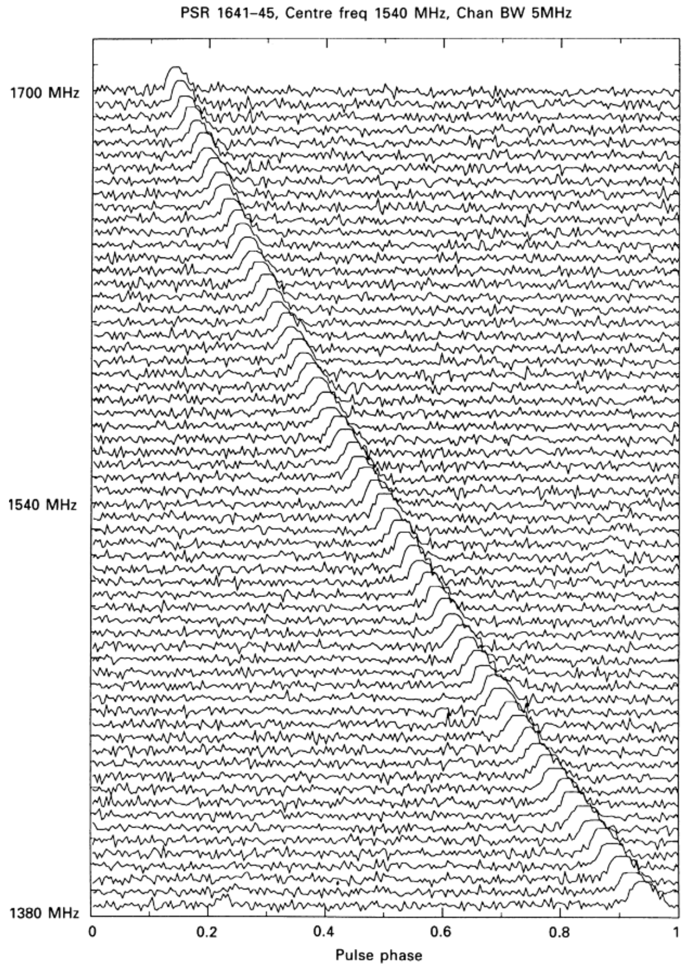

FIGURE 16.20 Dispersion of the pulse from PSR 1641-45. (Figure from Lyne and Graham-Smith, *Pulsar Astronomy*, ©Cambridge University Press, New York, 1990. Reprinted with the permission of Cambridge University Press.)

The Basic Pulsar Model

The basic pulsar model, shown in Fig. 16.25, consists of a rapidly rotating neutron star with a strong dipole magnetic field (two poles, north and south) that is inclined to the rotation axis at an angle θ. As explained in the previous section, the rapid rotation and the strong dipole field both arise naturally following the collapse of the core of a supergiant star.

First, we need to obtain a measure of the strength of the pulsar's magnetic field. As the pulsar rotates, the magnetic field at any point in space will change rapidly. According to Faraday's law, this will induce an electric field at that point. Far from the star (near the **light cylinder** defined in Fig. 16.26) the time-varying electric and magnetic fields form an electromagnetic wave that carries energy away from the star. For this particular situation,

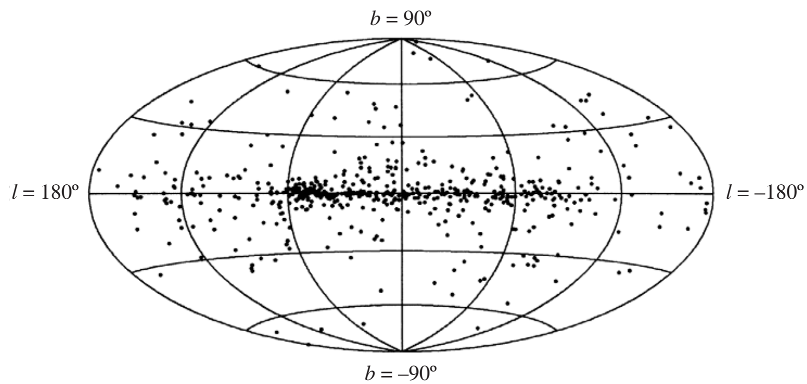

FIGURE 16.21 Distribution of 558 pulsars in galactic coordinates, with the center of the Milky Way in the middle. The clump of pulsars at $\ell = 60°$ is a selection effect due to the fixed orientation of the Arecibo radio telescope. (Figure from Taylor, Manchester, and Lyne, *Ap. J. Suppl.*, *88*, 529, 1993.)

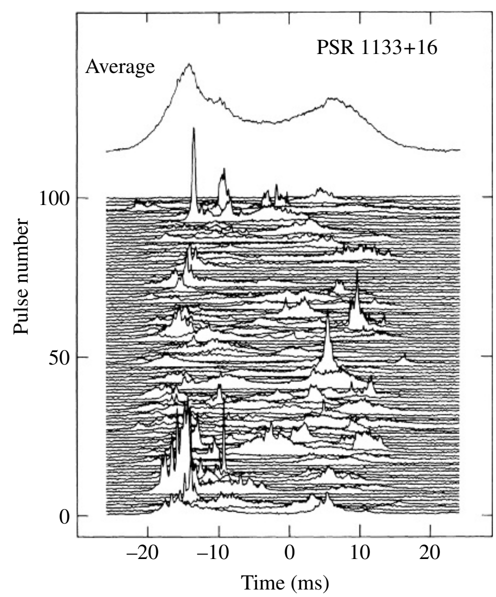

FIGURE 16.22 The average of 500 pulses (top) and a series of 100 consecutive pulses (below) for PSR 1133+16. (Figure adapted from Cordes, *Space Sci. Review*, *24*, 567, 1979.)

the radiation is called **magnetic dipole radiation**. Although it is beyond the scope of this book to consider the model in detail, we note that the energy per second emitted by the rotating magnetic dipole is

$$\boxed{\frac{dE}{dt} = -\frac{32\pi^5 B^2 R^6 \sin^2\theta}{3\mu_0 c^3 P^4},} \tag{16.31}$$

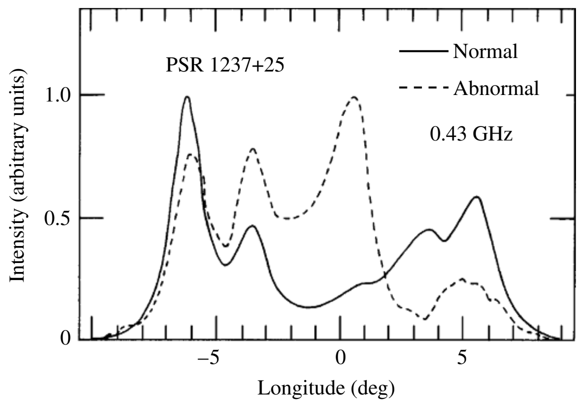

FIGURE 16.23 Changes in the integrated pulse profile of PSR 1237+25 due to mode switching. This pulsar displays five distinct subpulses. (Figure adapted from Bartel et al., *Ap. J.*, *258*, 776, 1982.)

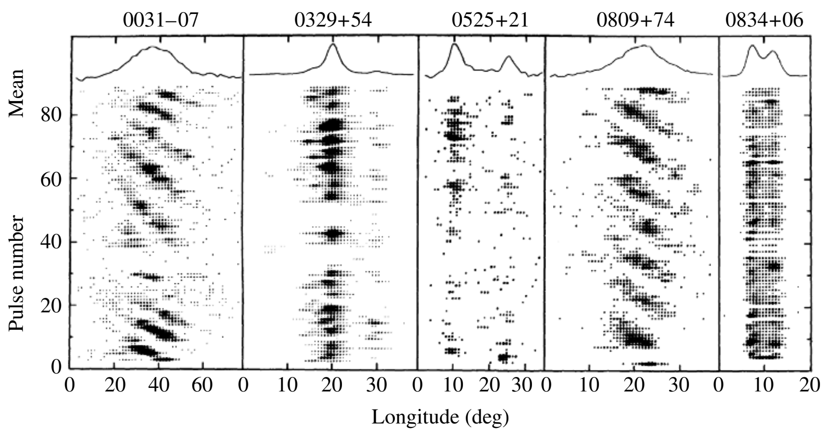

FIGURE 16.24 Drifting subpulses for two pulsars; note that PSR 0031-07 also nulls. (Figure from Taylor et al., *Ap. J.*, *195*, 513, 1975.)

where B is the field strength at the magnetic pole of the star of radius R. The minus sign indicates that the neutron star is drained of energy, causing its rotation period, P, to increase. Note that the factor of $1/P^4$ means that the neutron star will lose energy much more quickly at smaller periods. Since the average pulsar period is 0.79 s, most pulsars are born spinning considerably faster than their current rates, with typical initial periods of a few milliseconds.

Assuming that all of the rotational kinetic energy lost by the star is carried away by magnetic dipole radiation, $dE/dt = dK/dt$. Using Eqs. (16.30) and (16.31), this is

$$-\frac{32\pi^5 B^2 R^6 \sin^2\theta}{3\mu_0 c^3 P^4} = -\frac{4\pi^2 I \dot{P}}{P^3}. \tag{16.32}$$

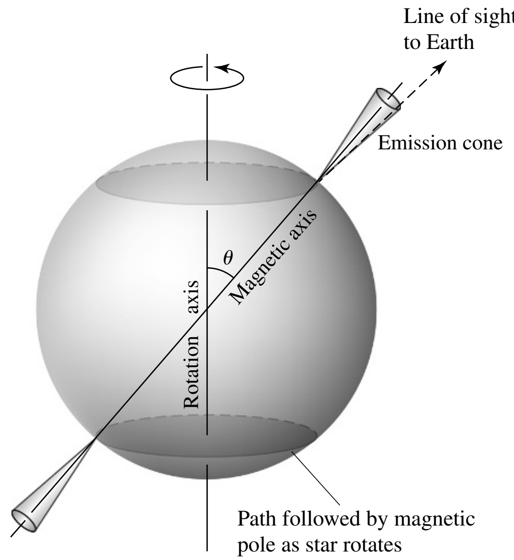

FIGURE 16.25 A basic pulsar model.

This can be easily solved for the magnetic field at the pole of the neutron star,

$$B = \frac{1}{2\pi R^3 \sin\theta}\sqrt{\frac{3\mu_0 c^3 I P \dot{P}}{2\pi}}. \tag{16.33}$$

Example 16.7.2. We will estimate the magnetic field strength at the poles of the Crab pulsar (PSR 0531-21), with $P = 0.0333$ s and $\dot{P} = 4.21 \times 10^{-13}$. Assuming that $\theta = 90°$, Eq. (16.33) then gives a value of 8.0×10^8 T. As we have seen, the Crab pulsar is interacting with the dust and gas in the surrounding nebula, so there are other torques that contribute to slowing down the pulsar's spin. This value of B is therefore an overestimate; the accepted value of the Crab pulsar's magnetic field is 4×10^8 T.[38] Values of B around 10^8 T are typical for most pulsars.

However, repeating the calculation for PSR 1937+214 with $P = 0.00156$ s, $\dot{P} = 1.05 \times 10^{-19}$, and assuming the same value for the moment of inertia, we find the magnetic field strength to be only $B = 8.6 \times 10^4$ T. This much smaller value distinguishes the millisecond pulsars and provides another hint that these fastest pulsars may have a different origin or environment.

Correlation Between Period Derivatives and Pulsar Classes

Figure 16.27 shows the distribution of period derivatives for pulsars as a function of pulsar period. Although the vast majority of pulsars fall into a large grouping in the middle of

[38]The suggestion that the Crab Nebula is powered by the magnetic dipole radiation from a rotating neutron star was made by the Italian astronomer Franco Pacini in 1967, a year *before* the discovery of pulsars!

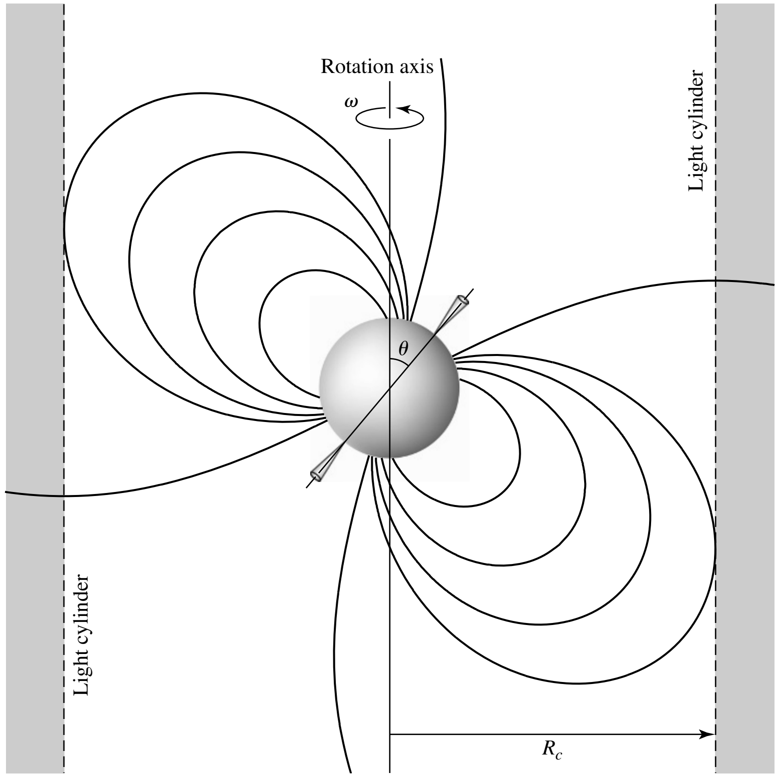

FIGURE 16.26 The light cylinder around a rotating neutron star. The cylinder's radius R_c is where a point co-rotating with the neutron star would move at the speed of light: $R_c = c/\omega = cP/2\pi$.

the plot, the millisecond pulsars show a clear correlation with pulsars known to exist in binary systems. Other classes of pulsars are also evident: Pulsars known to emit energy at X-ray wavelengths have the longest periods and have the largest period derivatives, whereas high-energy pulsars that emit energies from radio frequencies through the infrared or higher frequencies tend to have larger values of $\dot{P}$ but otherwise typical periods. Note that although nearly all of the pulsars represented in Fig. 16.27 have positive values of $\dot{P}$, some of them, primarily the binary pulsars, actually have values of $\dot{P} < 0$, meaning that their periods are decreasing (they are speeding up!). Figure 16.27 may be compared with the histogram of pulsar periods shown in Fig. 16.14.

Toward a Model of Pulsar Emission

Developing a detailed model of the pulsar's emission mechanism has been an exercise in frustration because almost every observation is open to more than one interpretation. The emission of radiation is the most poorly understood aspect of pulsars, and at present there is agreement only on the most general features of how a neutron star manages to produce radio waves. The following discussion summarizes a popular model of the pulse process.

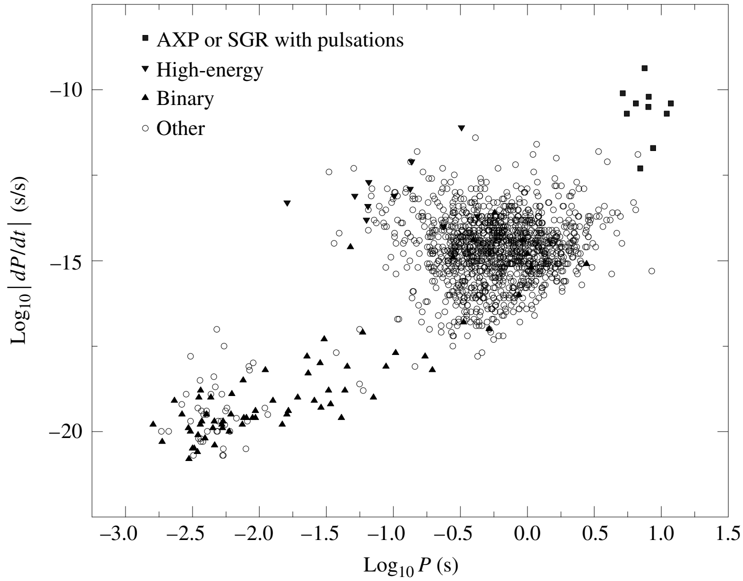

FIGURE 16.27 The absolute value of the time derivative of period ($|\dot{P}|$) versus period (P) for all pulsars for which $\dot{P}$ has been determined. Special classes of pulsars are depicted separately: Anomalous X-ray pulsars (AXP) or Soft Gamma Repeaters (SGR) with pulsations, high-energy pulsars with emitted frequencies between radio and infrared or higher, and binary pulsars (with one or more known binary companions) are depicted separately. All remaining pulsars are indicated as "other." Note the abundance of known binary pulsars among the millisecond pulsars. (Data from Manchester, Hobbs, Teoh, and Hobbs, *A. J.*, *129*, 1993, 2005. Data available at `http://www.atnf.csiro.au/research/pulsar/psrcat`.)

You should keep in mind, however, that there is as yet no general consensus on whether the object being discussed actually occurs in nature or only in the minds of astrophysicists!

It is at least certain that the rapidly changing magnetic field near the rotating pulsar induces a huge electric field at the surface. The electric field of about 6.3×10^{10} V m^{-1} easily overcomes the pull of gravity on charged particles in the neutron star's crust. For example, the electric force on a proton is about 300 million times stronger than the force of gravity, and the ratio of the electric force on an electron to the gravitational force is even more overwhelming. Depending on the direction of the electric field, either negatively charged electrons or positively charged ions will be continuously ripped from the neutron star's polar regions. This creates a **magnetosphere** of charged particles surrounding the pulsar that is dragged around with the pulsar's rotation. However, the speed of the co-rotating particles cannot exceed the speed of light, so at the light cylinder the charged particles are spun away, carrying the magnetic field with them in a pulsar "wind." Such a wind may be responsible for the replenishment of the Crab Nebula's magnetic field and the continual delivery of relativistic particles needed to keep the nebula shining.

The charged particles ejected from the vicinity of the pulsar's magnetic poles are quickly accelerated to relativistic speeds by the induced electric field. As the electrons follow the curved magnetic field lines, they emit curvature radiation in the form of energetic gamma-ray photons. This radiation is emitted in a narrow beam in the instantaneous direction of motion of the electron, a consequence of the relativistic headlight effect discussed in Section 4.3. Each gamma-ray photon has so much energy that it can spontaneously convert this energy into an electron–positron pair via Einstein's relation $E = mc^2$. (This process, described by $\gamma \rightarrow e^- + e^+$, is just the inverse of the annihilation process mentioned in Section 10.3 for the Sun's interior.) The electrons and positrons are accelerated and in turn emit their own gamma rays, which create more electron–positron pairs, and so on. A cascade of pair production is thus initiated near the magnetic poles of the neutron star. Coherent beams of curvature radiation emitted by bunches of these particles may be responsible for the individual subpulses that contribute to the integrated pulse profile.

As these particles continue to curve along the magnetic field lines, they emit a continuous spectrum of curvature radiation in the forward direction, producing a narrow cone of radio waves radiating from the magnetic polar regions.[39] As the neutron star rotates, these radio waves sweep through space in a way reminiscent of the light from a rotating lighthouse beacon. If the beam happens to fall on a radio telescope on a blue-green planet in a distant Solar System, the astronomers there will detect a regular series of brief radio pulses.

As the pulsar ages and slows down, the structure of the underlying neutron star must adapt to the reduced rotational stresses. As a consequence, perhaps the crust settles a fraction of a millimeter and the star spins faster as a result of its decreased moment of inertia, or perhaps the superfluid vortices in the neutron star's core become momentarily "unpinned" from the underside of the solid crust where they are normally attached, giving the crust a sudden jolt. Either possibility could produce a small but abrupt increase in the rotation speed, and the astronomers on Earth would record a glitch for the pulsar (recall Fig. 16.16).

The question of a pulsar's final fate, as its period increases beyond several seconds, has several possible answers. It may be that the neutron star's magnetic field, originally produced by the collapse of the pre-supernova star's degenerate stellar core, decays with a characteristic time of 9 million years or so. Then, at some future time when the pulsar's period has been reduced to several seconds, its magnetic field may no longer be strong enough to sustain the pulse mechanism, and the pulsar turns off. On the other hand, it may be that the magnetic field does not decay appreciably but is maintained by a dynamo-like mechanism involving the differential rotation of the crust and core of the neutron star. However, rotation itself is an essential ingredient of any pulsar emission mechanism. As a pulsar ages and slows down, its beam will become weaker even if the magnetic field does not decay. In this case, the radio pulses may become too faint to be detected as the pulsar simply fades below the sensitivity of radio telescopes. The timescale for the decay of a neutron star's magnetic field is a matter of considerable debate, and both scenarios are consistent with the observations.

Magnetars and Soft Gamma Repeaters

The preceding sketch reflects the current state of uncertainty about the true nature of pulsars. There are few objects in astronomy that offer such a wealth of intriguing observational detail

[39]The visible, X-ray, and gamma-ray pulses received from the Crab, Vela, Circinus, and Geminga pulsars may originate farther out in the pulsar's magnetosphere.

and yet are so lacking in a consistent theoretical description. Regardless of whether the basic picture outlined is vindicated or is supplanted by another view (perhaps involving a disk of material surrounding the neutron star), pulsar theorists will continue to take advantage of this unique natural laboratory for studying matter under the most extreme conditions.

To complicate the picture further, it is now believed that a class of extremely magnetic neutron stars known as **magnetars** exists. Magnetars have magnetic field strengths that are on the order of 10^{11} T, several orders of magnitude greater than typical pulsars. They also have relatively slow rotation periods of 5 to 8 seconds. Magnetars were first proposed to explain the **soft gamma repeaters** (SGRs), objects that emit bursts of hard X-rays and soft gamma-rays with energies of up to 100 keV (recall Fig. 16.27). Only a few SGRs are known to exist in the Milky Way Galaxy, and one has been detected in the Large Magellanic Cloud. Each of the SGRs is also known to correlate with supernova remnants of fairly young age ($\sim 10^4$ y). This would suggest that magnetars, if they are the source of the SGRs, are short-lived phenomena. Perhaps the Galaxy has many "extinct," or low-energy, magnetars scattered through it.

The emission mechanism of intense X-rays from SGRs is thought to be associated with stresses in the magnetic fields of magnetars that cause the surface of the neutron star to crack. The resulting readjustment of the surface produces a *super-Eddington* release of energy (roughly 10^3 to 10^4 times the Eddington luminosity limit in X-rays). In order to obtain such high luminosities, it is believed that the radiation must be confined; hence the need for very high magnetic field strengths.

Magnetars are distinguished from ordinary pulsars by the fact that the energy of the magnetar's field plays the major role in the energetics of the system, rather than rotation, as is the case for pulsars. Clearly much remains to be learned about the exotic environment of rapidly rotating, degenerate spheres with radii on the order of 10 km and densities exceeding the density of the nucleus of an atom.

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PROBLEMS

16.1 The most easily observed white dwarf in the sky is in the constellation of Eridanus (the River Eridanus). Three stars make up the 40 Eridani system: 40 Eri A is a 4th-magnitude star similar to the Sun; 40 Eri B is a 10th-magnitude white dwarf; and 40 Eri C is an 11th-magnitude red M5 star. This problem deals only with the latter two stars, which are separated from 40 Eri A by 400 AU.

(a) The period of the 40 Eri B and C system is 247.9 years. The system's measured trigonometric parallax is $0.201''$ and the true angular extent of the semimajor axis of the reduced mass is $6.89''$. The ratio of the distances of 40 Eri B and C from the center of mass is $a_B/a_C = 0.37$. Find the mass of 40 Eri B and C in terms of the mass of the Sun.

(b) The absolute bolometric magnitude of 40 Eri B is 9.6. Determine its luminosity in terms of the luminosity of the Sun.

(c) The effective temperature of 40 Eri B is 16,900 K. Calculate its radius, and compare your answer to the radii of the Sun, Earth, and Sirius B.

(d) Calculate the average density of 40 Eri B, and compare your result with the average density of Sirius B. Which is more dense, and why?

(e) Calculate the product of the mass and volume of both 40 Eri B and Sirius B. Is there a departure from the mass–volume relation? What might be the cause?

16.2 The helium absorption lines seen in the spectra of DB white dwarfs are formed by excited He I atoms with one electron in the lowest ($n = 1$) orbital and the other in an $n = 2$ orbital. White dwarfs of spectral type DB are not observed with temperatures below about 11,000 K. Using what you know about spectral line formation, give a *qualitative* explanation why the helium lines would not be seen at lower temperatures. As a DB white dwarf cools below 12,000 K, into what spectral type does it change?

16.3 Deduce a rough upper limit for X, the mass fraction of hydrogen, in the interior of a white dwarf. *Hint:* Use the mass and average density for Sirius B in the equations for the nuclear energy generation rate, and take $T = 10^7$ K for the central temperature. Set ψ_{pp} and $f_{pp} = 1$ in Eq. (10.47) for the pp chain, and $X_{\rm CNO} = 1$ in Eq. (10.59) for the CNO cycle.

16.4 Estimate the ideal gas pressure and the radiation pressure at the center of Sirius B, using 3×10^7 K for the central temperature. Compare these values with the estimated central pressure, Eq. (16.1).

16.5 By equating the pressure of an ideal gas of electrons to the pressure of a degenerate electron gas, determine a condition for the electrons to be degenerate, and compare it with the condition of Eq. (16.6). Use the exact expression (Eq. 16.12) for the electron degeneracy pressure.

16.6 In the extreme relativistic limit, the electron speed $v = c$ must be used instead of Eq. (16.10) to find the electron degeneracy pressure. Use this to repeat the derivation of Eq. (16.11) and find

$$P \approx \frac{\hbar c}{\sqrt{3}} \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{4/3}.$$

16.7 **(a)** At what speed do relativistic effects become important at a level of 10%? In other words, for what value of v does the Lorentz factor, γ, become equal to 1.1?

(b) Estimate the density of the white dwarf for which the speed of a degenerate electron is equal to the value found in part (a).

(c) Use the mass–volume relation to find the approximate mass of a white dwarf with this average density. This is roughly the mass where white dwarfs depart from the mass–volume relation.

16.8 Crystallization will occur in a cooling white dwarf when the electrostatic potential energy between neighboring nuclei, $Z^2e^2/4\pi\epsilon_0 r$, dominates the characteristic thermal energy kT.

The ratio of the two is defined to be Γ,

$$\Gamma = \frac{Z^2 e^2}{4\pi \epsilon_0 r k T}.$$

In this expression, the distance r between neighboring nuclei is customarily (and somewhat awkwardly) defined to be the radius of a sphere whose volume is equal to the volume per nucleus. Specifically, since the average volume per nucleus is Am_H/ρ, r is found from

$$\frac{4}{3}\pi r^3 = \frac{Am_H}{\rho}.$$

(a) Calculate the value of the average separation r for a 0.6 $M_\odot$ pure carbon white dwarf of radius 0.012 $R_\odot$.

(b) Much effort has been spent on precise numerical calculations of Γ to obtain increasingly realistic cooling curves. The results indicate a value of about $\Gamma = 160$ for the onset of crystallization. Estimate the interior temperature, T_c, at which this occurs.

(c) Estimate the luminosity of a pure carbon white dwarf with this interior temperature. Assume a composition like that of Example 16.5.1 for the nondegenerate envelope.

(d) For roughly how many years could the white dwarf sustain the luminosity found in part (c), using just the latent heat of kT per nucleus released upon crystallization? Compare this amount of time (when the white dwarf cools more slowly) with Fig. 16.9.

16.9 In the *liquid-drop model* of an atomic nucleus, a nucleus with mass number A has a radius of $r_0 A^{1/3}$, where $r_0 = 1.2 \times 10^{-15}$ m. Find the density of this nuclear model.

16.10 If our Moon were as dense as a neutron star, what would its diameter be?

16.11 **(a)** Consider two point masses, each having mass m, that are separated vertically by a distance of 1 cm just above the surface of a neutron star of radius R and mass M. Using Newton's law of gravity (Eq. 2.11), find an expression for the ratio of the gravitational force on the lower mass to that on the upper mass, and evaluate this expression for $R = 10$ km, $M = 1.4\ M_\odot$, and $m = 1$ g.

(b) An iron cube 1 cm on each side is held just above the surface of the neutron star described in part (a). The density of iron is 7860 kg m^{-3}. If iron experiences a stress (force per cross-sectional area) of 4.2×10^7 N m^{-2}, it will be permanently stretched; if the stress reaches 1.5×10^8 N m^{-2}, the iron will rupture. What will happen to the iron cube? (*Hint:* Imagine concentrating half of the cube's mass on each of its top and bottom surfaces.) What would happen to an iron meteoroid falling toward the surface of a neutron star?

16.12 Estimate the neutron degeneracy pressure at the center of a 1.4 $M_\odot$ neutron star (take the central density to be 1.5×10^{18} kg m^{-3}), and compare this with the estimated pressure at the center of Sirius B.

16.13 **(a)** Assume that at a density just below neutron drip, all of the neutrons are in heavy neutron-rich nuclei such as $^{118}_{36}\mathrm{Kr}$. Estimate the pressure due to relativistic degenerate electrons.

(b) Assume (*wrongly!*) that at a density just above neutron drip, all of the neutrons are free (and not in nuclei). Estimate the speed of the degenerate neutrons and the pressure they would produce.

16.14 Suppose that the Sun were to collapse down to the size of a neutron star (10-km radius).

(a) Assuming that no mass is lost in the collapse, find the rotation period of the neutron star.

(b) Find the magnetic field strength of the neutron star.

Even though our Sun will not end its life as a neutron star, this shows that the conservation of angular momentum and magnetic flux can easily produce pulsar-like rotation speeds and magnetic fields.

16.15 **(a)** Use Eq. (14.14) with $\gamma = 5/3$ to calculate the fundamental radial pulsation period for a one-zone model of a pulsating white dwarf (use the values for Sirius B) and a 1.4 $M_\odot$ neutron star. Compare these to the observed range of pulsar periods.

(b) Use Eq. (16.29) to calculate the minimum rotation period for the same stars, and compare them to the range of pulsar periods.

(c) Give an explanation for the similarity of your results.

16.16 **(a)** Determine the minimum rotation period for a 1.4 $M_\odot$ neutron star (the fastest it can spin without flying apart). For convenience, assume that the star remains spherical with a radius of 10 km.

(b) Newton studied the equatorial bulge of a homogeneous fluid body of mass M that is *slowly* rotating with angular velocity Ω. He proved that the difference between its equatorial radius (E) and its polar radius (P) is related to its average radius (R) by

$$\frac{E - P}{R} = \frac{5\Omega^2 R^3}{4GM}.$$

Use this to estimate the equatorial and polar radii for a 1.4 $M_\odot$ neutron star rotating with twice the minimum rotation period you found in part (a).

16.17 If you measured the period of PRS 1937+214 and obtained the value on page 588, about how long would you have to wait before the last digit changed from a "5" to a "6"?

16.18 Consider a pulsar that has a period P_0 and period derivative $\dot{P}_0$ at $t = 0$. Assume that the product $P\dot{P}$ remains constant for the pulsar (cf. Eq. 16.32).

(a) Integrate to obtain an expression for the pulsar's period P at time t.

(b) Imagine that you have constructed a clock that would keep time by counting the radio pulses received from this pulsar. Suppose you also have a *perfect* clock ($\dot{P} = 0$) that is initially synchronized with the pulsar clock when they both read zero. Show that when the perfect clock displays the characteristic lifetime $P_0/\dot{P}_0$, the time displayed by the pulsar clock is $(\sqrt{3} - 1)P_0/\dot{P}_0$.

16.19 During a glitch, the period of the Crab pulsar decreased by $|\Delta P| \approx 10^{-8}P$. If the increased rotation was due to an overall contraction of the neutron star, find the change in the star's radius. Assume that the pulsar is a rotating sphere of uniform density with an initial radius of 10 km.

16.20 The Geminga pulsar has a period of $P = 0.237$ s and a period derivative of $\dot{P} = 1.1 \times 10^{-14}$. Assuming that $\theta = 90°$, estimate the magnetic field strength at the pulsar's poles.

16.21 **(a)** Find the radii of the light cylinders for the Crab pulsar and for the slowest pulsar, PSR 1841-0456. Compare these values to the radius of a 1.4 $M_\odot$ neutron star.

(b) The strength of a magnetic dipole is proportional to $1/r^3$. Determine the ratio of the magnetic field strengths at the light cylinder for the Crab pulsar and for PSR 1841-0456.

16.22 **(a)** Integrate Eq. (16.32) to obtain an expression for a pulsar's period P at time t if its initial period was P_0 at time $t = 0$.

(b) Assuming that the pulsar has had time to slow down enough that $P_0 \ll P$, show that the age t of the pulsar is given approximately by

$$t = \frac{P}{2\dot{P}},$$

where $\dot{P}$ is the period derivative at time t.

(c) Evaluate this age for the case of the Crab pulsar, using the values found in Example 16.7.1. Compare your answer with the known age.

16.23 One way of qualitatively understanding the flow of charged particles into a pulsar's magnetosphere is to imagine a charged particle of mass m and charge e (the fundamental unit of charge) at the equator of the neutron star. Assume for convenience that the star's rotation carries the charge perpendicular to the pulsar's magnetic field. The moving charge experiences a magnetic Lorentz force of $F_m = evB$ and a gravitational force, F_g. Show that the ratio of these forces is

$$\frac{F_m}{F_g} = \frac{2\pi eBR}{Pmg},$$

where R is the star's radius and g is the acceleration due to gravity at the surface. Evaluate this ratio for the case of a proton at the surface of the Crab pulsar, using a magnetic field strength of 10^8 T.

16.24 Find the minimum photon energy required for the creation of an electron–positron pair via the pair-production process $\gamma \rightarrow e^- + e^+$. What is the wavelength of this photon? In what region of the electromagnetic spectrum is this wavelength found?

16.25 A subpulse involves a very narrow radio beam with a width between 1° and 3°. Use Eq. (4.43) for the headlight effect to calculate the minimum speed of the electrons responsible for a 1° subpulse.

CHAPTER 17 General Relativity and Black Holes

17.1 ■ THE GENERAL THEORY OF RELATIVITY

Gravity, the weakest of the four forces of nature, plays a fundamental role in sculpting the universe on the largest scale. Newton's law of universal gravitation,

$$F = G\frac{Mm}{r^2}, \tag{17.1}$$

remained an unquestioned cornerstone of astronomers' understanding of heavenly motions until the beginning of the twentieth century. Its application had explained the motions of the known planets and had accurately predicted the existence and position of the planet Neptune in 1846. The sole blemish on Newtonian gravitation was the inexplicably large rate of shift in the orientation of Mercury's orbit.

The gravitational influences of the other planets cause the major axis of Mercury's elliptical orbit to slowly swing around the Sun in a counterclockwise direction relative to the fixed stars; see Fig. 17.1. The angular position at which perihelion occurs shifts at a rate of $574''$ per century.[1] However, Newton's law of gravity was unable to explain $43''$ per century of this shift, an inconsistency that led some mid-nineteenth century physicists to suggest that Eq. (17.1) should be modified from an exact inverse-square law. Others thought that an unseen planet, nicknamed Vulcan, might occupy an orbit inside Mercury's.

The Curvature of Spacetime

Between the years 1907 and 1915, Albert Einstein developed a new theory of gravity, his **general theory of relativity**. In addition to resolving the mystery of Mercury's orbit, it predicted many new phenomena that were later confirmed by experiment. In this and the next section we will describe just enough of the physical content of general relativity to provide the background needed for future discussions of black holes and cosmology.

[1]The value of 1.5° per century encountered in some texts includes the very large effect of the precession of Earth's rotation axis on the celestial coordinate system, described in Section 1.3.

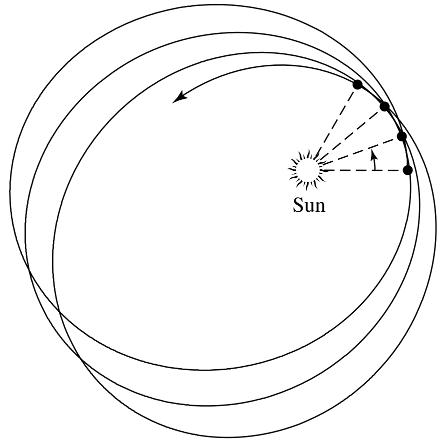


FIGURE 17.1 The perihelion shift of Mercury's orbit. Both the eccentricity of the orbit and the amount of shift in the location of perihelion in successive orbits have been exaggerated to better show the effect.

Einstein's view of the universe provides an exhilarating challenge to the imaginations of all students of astrophysics. But, before embarking on our study of general relativity, it will be helpful to take an advanced look at this new gravitational landscape.

The general theory of relativity is fundamentally a geometric description of how distances (intervals) in spacetime are measured in the presence of mass. For the moment, the effects on space and time will be considered separately, although you should always keep in mind that relativity deals with a unified spacetime. Near an object, both space and time must be described in a new way.

Distances between points in the space surrounding a massive object are altered in a way that can be interpreted as space becoming *curved* through a fourth spatial dimension perpendicular to *all* of the usual three spatial directions. The human mind balks at picturing this situation, but an analogy is easily found. Imagine four people holding the corners of a rubber sheet, stretching it tight and flat. This represents the flatness of empty space that exists in the absence of mass. Also imagine that a polar coordinate system has been painted on the sheet, with evenly spaced concentric circles spreading out from its center. Now lay a heavy bowling ball (representing the Sun) at the center of the sheet, and watch the indentation of the sheet as it curves down and stretches in response to the ball's weight, as pictured in Fig. 17.2. Closer to the ball, the sheet's curvature increases and the distance between points on the circles is stretched more. Just as the sheet curves in a third direction perpendicular to its original flat two-dimensional plane, the space surrounding a massive object may be thought of as curving in a fourth spatial dimension perpendicular to the usual three of "flat space."[2] The fact that mass has an effect on the surrounding space is the first essential element of general relativity. The curvature of space is just one aspect of the effect

[2]It is important to note that this fourth spatial dimension has nothing at all to do with the role played by time as a fourth nonspatial *coordinate* in the theory of relativity.

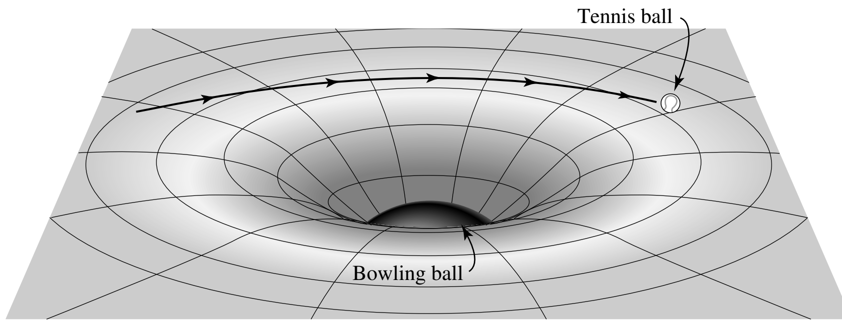


FIGURE 17.2 Rubber sheet analogy for curved space around the Sun. It is assumed that the rubber sheet is much larger than the area of curvature, so that the edges of the sheet have no effect on the curvature produced by the central mass.

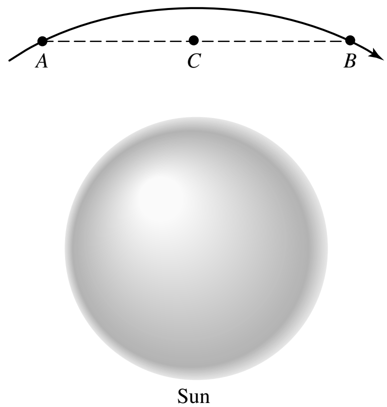


FIGURE 17.3 A photon's path around the Sun is shown by the solid line. The bend in the photon's trajectory is greatly exaggerated.

of mass on spacetime. In the language of unified spacetime, *mass acts on spacetime, telling it how to curve*.

Now imagine rolling a tennis ball, representing a planet, across the sheet. As it passes near the bowling ball, the tennis ball's path is curved. If the ball were rolled in just the right way under ideal conditions, it could even "orbit" the more massive bowling ball. In a similar manner, a planet orbits the Sun as it responds to the curved spacetime around it. Thus *curved spacetime acts on mass, telling it how to move*.

The passage of a ray of light near the Sun can be represented by rolling a ping-pong ball very rapidly past the bowling ball. Although the analogy with a massless photon is strained, it is reasonable to expect that as the photon moves through the curved space surrounding the Sun, its path will be deflected from a straight line. The bend of the photon's trajectory is small because the photon's speed carries it quickly through the curved space; see Fig. 17.3. In general relativity, gravity is the result of objects moving through curved spacetime, and everything that passes through, even massless particles such as photons, is affected.

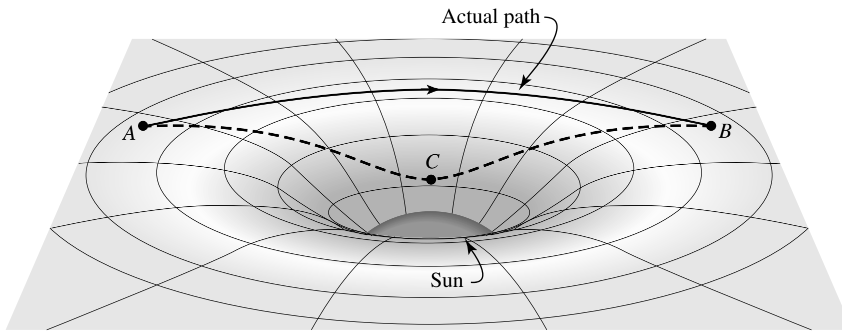


FIGURE 17.4 Comparison of two photon paths through curved space between points A and B. The projection of the path ACB onto the plane is the straight line depicted in Fig. 17.3.

Figure 17.3 hints at another aspect of general relativity. Since nothing can move between two points in space faster than light, light must always follow the quickest route between any two points.[3] In flat, empty space, this path is a straight line, but what is the quickest route through curved space? Suppose we use a series of mirrors to force the light beam to travel between points A and B by the apparent "shortcut" indicated by the dashed lines in Figs. 17.3 and 17.4. Would the light taking the dashed path outrace the beam free to follow its natural route through curved space? The answer is no—the curved beam would win the race. This result seems to imply that the beam following the dashed line would slow down along the way. However, this inference can't be correct because, according to the postulates of relativity (page 88), every observer, including one at point C, measures the same value for the speed of light. There are just two possible answers. The distance along the dashed line might actually be *longer* than the light beam's natural path, and/or time might run more slowly along the dashed path; either would retard the beam's passage. In fact, according to general relativity, these effects contribute equally to delaying the light beam's trip from A to B along the dashed line. The curving light beam actually does travel the *shorter* path. If two space travelers were to lay meter sticks end-to-end along the two paths, the dashed path would require a greater number of meter sticks because it penetrates farther into curved space, as shown in Fig. 17.4. In addition, the curvature of space involves a concomitant slowing down of time, so clocks placed along the dashed path would actually *run more slowly*. This is the final essential feature of general relativity: *Time runs more slowly in curved spacetime*.

It is important to note that all of the foregoing ideas have been tested experimentally many times, and in every case the results agree with general relativity. As soon as Einstein completed his theory, he applied it to the problem of Mercury's unexplained residual perihelion shift of 43″ per century. Einstein wrote that his heart raced when his calculations exactly explained the discrepancy in terms of the planet's passage through the curved space near the Sun, saying that, "For a few days, I was beside myself with joyous excitement." Another triumph came in 1919 when the curving path of starlight passing near the Sun was first measured, by Arthur Stanley Eddington, during a total solar eclipse. As shown

[3]Throughout this chapter, light is assumed to be traveling in a vacuum.

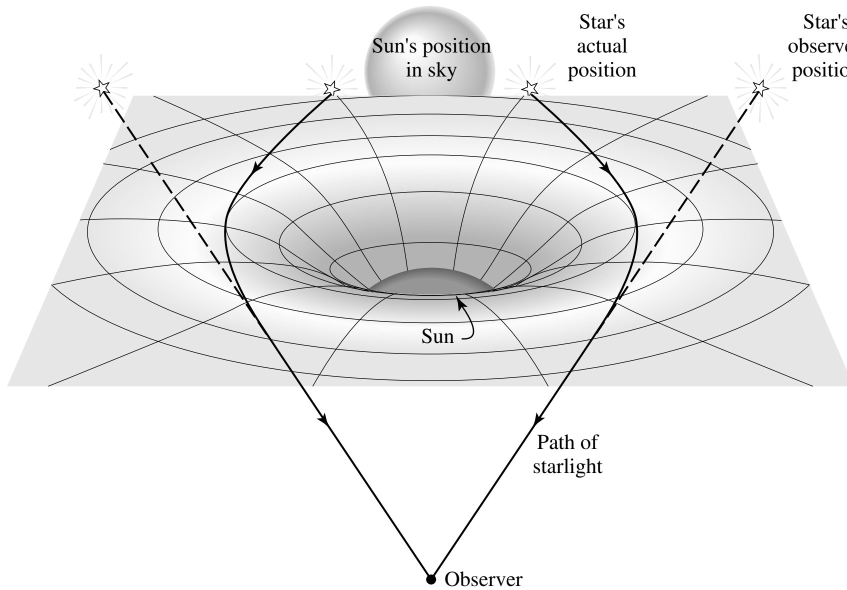


FIGURE 17.5 Bending of starlight measured during a solar eclipse.

in Fig. 17.5, the apparent positions of stars close to the Sun's eclipsed edge were shifted from their actual positions by a small angle. Einstein's theory predicted that this angular deflection would be 1.75″, in good agreement with Eddington's observations. General relativity has been tested continuously ever since. For instance, the superior conjunction of Mars that occurred in 1976 led to a spectacular confirmation of Einstein's theory. Radio signals beamed to Earth from the Viking spacecraft on Mars's surface were delayed as they traveled deep into the curved space surrounding the Sun. The time delay agreed with the predictions of general relativity to within 0.1%.

The Principle of Equivalence

It is now time to retrace our steps and discover how Einstein came to his revolutionary understanding of gravity as geometry. One of the postulates of special relativity states that the laws of physics are the same in all inertial reference frames. Accelerating frames of reference are not inertial frames, because they introduce fictitious forces that depend on the acceleration. For example, an apple at rest on the seat of a car will not remain at rest if the car suddenly brakes to a halt. However, the acceleration produced by the force of gravity has a unique aspect. This may be clearly seen by noting a fundamental difference between Newton's law of gravity and Coulomb's law for the electrical force (Eq. 5.9).

Consider two objects separated by a distance r, one of mass m and charge q, and the other of mass M and charge Q. The magnitude of the acceleration (a_g) of mass m *due to the gravitational force* is found from

$$ma_g = G\frac{mM}{r^2}, \tag{17.2}$$

while the magnitude of the acceleration (a_e) *due to the electrical force* is found from

$$ma_e = \frac{qQ}{4\pi\epsilon_0 r^2}. \tag{17.3}$$

The mass m on the left-hand sides is an *inertial mass* and measures the object's resistance to being accelerated (its inertia). On the right-hand sides, the masses m and M and charges q and Q are numbers that couple the masses or charges to their respective forces and determine the strength of these forces. The mystery is the appearance of m on both sides of the gravitational formula.

Why should a quantity that measures an object's inertia (which exists even in the complete absence of gravity) be the same as the "gravitational charge" that determines the force of gravity? The answer is that the notation in Eq. (17.2) is flawed, and the expression should properly be written as

$$m_i a_g = G\frac{m_g M_g}{r^2}$$

or

$$a_g = G\frac{M_g}{r^2}\frac{m_g}{m_i} \tag{17.4}$$

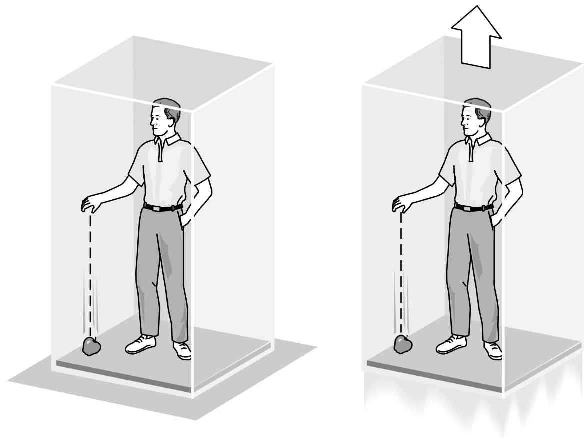
to clearly distinguish between the inertial and gravitational mass of each object. Similarly, for Eq. (17.3),

$$a_e = \frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}\frac{1}{m_i}.$$

In this case the only mass that enters the expression is the inertial mass.

It is an experimental fact, tested to a precision of 1 part in 10^{12}, that m_g/m_i in Eq. (17.4) is a constant. For convenience, this constant is chosen to be unity so the two types of mass will be numerically equal; if the gravitational mass were chosen to be *twice* the inertial mass, for example, the laws of physics would be unchanged except the gravitational constant G would be assigned a new value only one-fourth as large. The proportionality of the inertial and gravitational masses means that at a given location, all objects experience the *same* gravitational acceleration. The constancy of m_g/m_i is sometimes referred to as the **weak equivalence principle**.

This distinctive aspect of gravity, that every object falls with the same acceleration, has been known since the time of Galileo. It presented Einstein with both a problem and an opportunity to extend his theory of special relativity. He realized that if an entire laboratory were in free-fall, with all of its contents falling together, there would then be no way to detect its acceleration. In such a freely falling laboratory, it would be impossible to experimentally determine whether the laboratory was floating in space, far from any massive object, or falling freely in a gravitational field. Similarly, an observer watching an apple falling with an acceleration g toward the floor of a laboratory would be unable to tell whether the laboratory was on Earth or far out in space, accelerating at a rate g in the direction of the ceiling, as illustrated in Fig. 17.6. This posed a serious problem for the theory of special



(a) (b)

FIGURE 17.6 Gravity is equivalent to an accelerating laboratory: (a) a laboratory on Earth, and (b) a laboratory accelerating in space.

relativity, which requires that inertial reference frames have a constant velocity. Because gravity is equivalent to an accelerating laboratory, an inertial reference frame cannot even be defined in the presence of gravity. Einstein had to find a way to remove gravity from the laboratory.

In 1907, Einstein had "the happiest thought of my life."

> I was sitting in a chair in the patent office at Bern when all of a sudden a thought occurred to me: "If a person falls freely he will not feel his own weight." I was startled. This simple thought made a deep impression on me. It impelled me toward a theory of gravitation.

The way to eliminate gravity in a laboratory is to surrender to it by entering into a state of free-fall; see Fig. 17.7.[4] However, there was an obstacle to applying this to special relativity because its inertial reference frames are *infinite* collections of meter sticks and synchronized clocks (recall Section 4.1). It would be impossible to eliminate gravity everywhere in an infinite, freely falling reference frame, because different points would have to be falling at different rates in different directions (toward the center of Earth, for example). Einstein realized that he would have to use *local* reference frames, just small enough that the acceleration due to gravity would be essentially constant in both magnitude and direction everywhere inside the reference frame (see Fig. 17.8). Gravity would then be abolished inside a local, freely falling reference frame.

[4] *Free-fall* means that there are no nongravitational forces accelerating the laboratory. In his meditation on general relativity, *A Journey into Gravity and Spacetime* (see Suggested Readings), John A. Wheeler prefers the term *free-float*. Since gravity has been abolished, why should falling even be mentioned? You are also urged to browse through the pages of *Gravitation* by Misner, Thorne, and Wheeler (1973) for additional insights into general relativity.

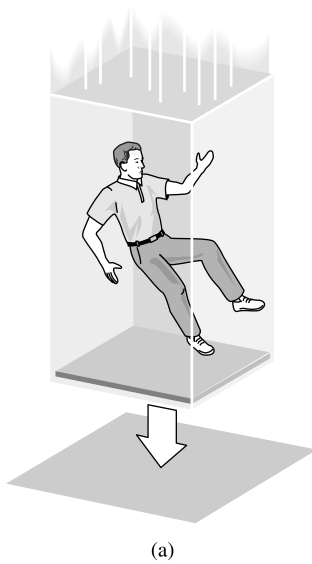


FIGURE 17.7 Gravity abolished in a freely falling laboratory: (a) a laboratory in free-fall, and (b) a laboratory floating in space.

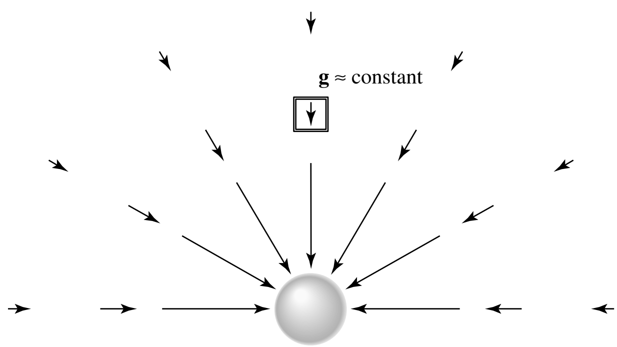


FIGURE 17.8 A local inertial reference frame, with $\mathbf{g} \approx$ constant inside. The arrows denote the gravitational acceleration vectors at those points around the mass.

In 1907 Einstein adopted this as the cornerstone of his theory of gravity, calling it the principle of equivalence.

> **The Principle of Equivalence:** All local, freely falling, nonrotating laboratories are fully equivalent for the performance of all physical experiments.

The restriction to nonrotating labs is necessary to eliminate the fictitious forces associated with rotation, such as the Coriolis and centrifugal forces. We will call these local, freely falling, nonrotating laboratories **local inertial reference frames**.

Note that special relativity is incorporated into the principle of equivalence. For example, measurements made from two local inertial frames in relative motion are related by the

Lorentz transformations (Eqs. 4.16–4.19) using the *instantaneous* value of the relative velocity between the two frames. Thus general relativity is in fact an extension of the theory of special relativity.

The Bending of Light

We now move on to two simple thought experiments involving the equivalence principle that demonstrate the curvature of spacetime. For the first experiment, imagine a laboratory suspended above the ground by a cable [see Fig. 17.9(a)]. Let a photon of light leave a horizontal flashlight at the same instant the cable holding the lab is severed [Fig. 17.9(b)]. Gravity has been abolished from this freely falling lab, so it is now a local inertial reference frame. According to the equivalence principle, an observer falling with the lab will measure the light's path across the room as a straight horizontal line, in agreement with all of the laws of physics. But another observer on the ground sees a lab that is falling under the influence of gravity. Because the photon maintains a constant height above the lab's floor, the ground observer must measure a photon that falls with the lab, following a curved path. This displays the spacetime curvature represented by the rubber sheet analogy. The curved path taken by the photon is the quickest route possible through the curved spacetime surrounding Earth.

The angle of deflection, ϕ, of the photon is very slight, as the following bit of geometry shows. Although the photon does not follow a circular path, we will use the *best-fitting circle* of radius r_c to the actual path measured by the ground observer. Referring to Fig. 17.10, the center of the best-fitting circle is at point O, and the arc of the circle subtends an angle ϕ (exaggerated in the figure) between the radii OA and OB. If the width of the lab is ℓ, then the photon crosses the lab in time $t = \ell/c$. (The difference between the length of the arc and the width of the lab is negligible.) In this amount of time, the lab falls a distance

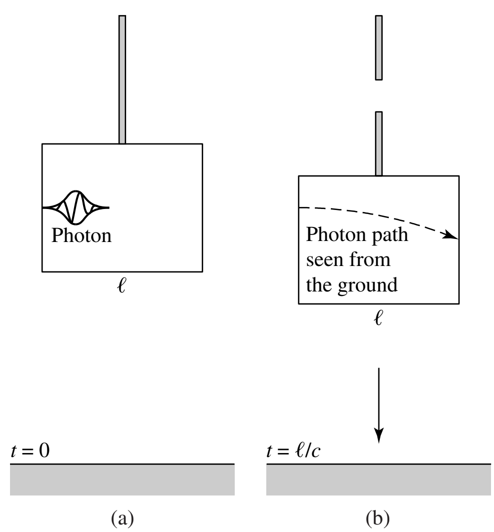


FIGURE 17.9 The equivalence principle for a horizontally traveling photon. The photon (a) leaves the left wall at $t = 0$, and (b) arrives at the right wall at $t = \ell/c$.

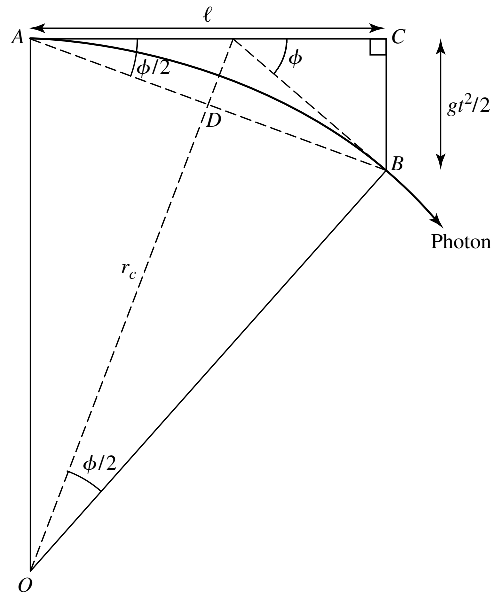


FIGURE 17.10 Geometry for the radius of curvature, r_c, and angular deflection, ϕ.

$d = \frac{1}{2}gt^2$. Because triangles ABC and OBD are similar (each containing a right angle and another angle $\phi/2$),

$$\overline{BC}/\overline{AC} = \overline{BD}/\overline{OD}$$

$$\left(\frac{1}{2}gt^2\right) \Big/ \ell = \left[\frac{\ell}{2\cos(\phi/2)}\right] \Big/ \overline{OD}.$$

In fact, ϕ is so small that we can set $\cos(\phi/2) \simeq 1$ and the distance $\overline{OD} \simeq r_c$. Then, using $t = \ell/c$ and $g = 9.8 \text{ m s}^{-2}$ for the acceleration of gravity near the surface of Earth, we find

$$r_c = \frac{c^2}{g} = 9.17 \times 10^{15} \text{ m}, \tag{17.5}$$

for the radius of curvature of the photon's path, which is nearly a light-year!

Of course, the angular deflection ϕ depends on the width ℓ of the lab. For example, if $\ell = 10$ m, then

$$\phi = \frac{\ell}{r_c} = 1.09 \times 10^{-15} \text{ rad},$$

or only 2.25×10^{-10} arcsecond. The large radius of the photon's path indicates that spacetime near Earth is only slightly curved. Nonetheless, the curvature is great enough to produce the circular orbits of satellites, which move slowly through the curved spacetime (slowly, that is, compared to the speed of light).

Gravitational Redshift and Time Dilation

Our second thought experiment also begins with the laboratory suspended above the ground by a cable. This time, monochromatic light of frequency ν_0 leaves a vertical flashlight on the floor at the same instant the cable holding the lab is severed. The freely falling lab is again a local inertial frame where gravity has been abolished, and so the equivalence principle requires that a frequency meter in the lab's ceiling record the *same* frequency, ν_0, for the light that it receives. But an observer on the ground sees a lab that is falling under the influence of gravity. As shown in Fig. 17.11, if the light has traveled upward a height h toward the meter in time $t = h/c$, then the meter has gained a downward speed toward the light of $v = gt = gh/c$ since the cable was released. Accordingly, we would expect that from the point of view of the ground observer, the meter should have measured a blueshifted frequency greater than ν_0 by an amount given by Eq. (4.33). For the slow free-fall speeds involved here, this expected *increase* in frequency is

$$\frac{\Delta\nu}{\nu_0} = \frac{v}{c} = \frac{gh}{c^2}.$$

But in fact, the meter recorded *no change* in frequency. Therefore there must be another effect of the light's upward journey through the curved spacetime around Earth that exactly compensates for this blueshift. This is a **gravitational redshift** that tends to *decrease* the frequency of the light as it travels upward a distance h, given by

$$\frac{\Delta\nu}{\nu_0} = -\frac{v}{c} = -\frac{gh}{c^2}. \tag{17.6}$$

An outside observer, not in free-fall inside the lab, would measure only this gravitational

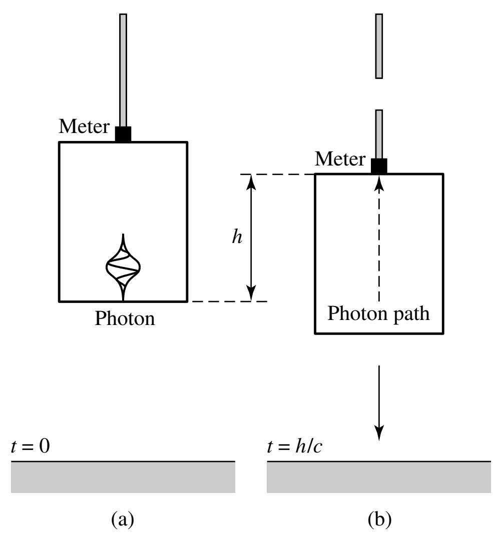


FIGURE 17.11 Equivalence principle for a vertically traveling light. The photon (a) leaves the floor at $t = 0$, and (b) arrives at the ceiling at $t = h/c$.

redshift. If the light were traveling downward, a corresponding blueshift would be measured. It is left as an exercise to show that this formula remains valid even if the light is traveling at an angle to the vertical, as long as h is taken to be the *vertical* distance covered by the light.

Example 17.1.1. In 1960, a test of the gravitational redshift formula was carried out at Harvard University. A gamma ray was emitted by an unstable isotope of iron, $^{57}_{26}\mathrm{Fe}$, at the bottom of a tower 22.6 m tall, and received at the top of the tower. Using this value for h, the expected decrease in frequency of the gamma ray due to the gravitational redshift is

$$\frac{\Delta\nu}{\nu_0} = -\frac{gh}{c^2} = -2.46 \times 10^{-15}, \tag{17.7}$$

in excellent agreement with the experimental result of $\Delta\nu/\nu_0 = -(2.57 \pm 0.26) \times 10^{-15}$. More precise experiments carried out since that time have obtained agreement to within 0.007%.

In actuality, the experiment was performed with both upward- and downward-traveling gamma rays, providing tests of both the gravitational redshift and blueshift.

An approximate expression for the total gravitational redshift for a beam of light that escapes out to infinity can be calculated by integrating Eq. (17.6) from an initial position r_0 to infinity, using $g = GM/r^2$ (Newtonian gravity) and setting h equal to the differential radial element, dr for a spherical mass, M, located at the origin. Some care must be taken when carrying out the integration, because Eq. (17.7) was derived using a *local* inertial reference frame. By integrating, we are really adding up the redshifts obtained for a chain of *different* frames. The radial coordinate r can be used to measure distances for these frames only if spacetime is nearly flat [that is, if the radius of curvature given by Eq. (17.5) is very large compared with r_0]. In this case, the "stretching" of distances seen previously in the rubber sheet analogy is not too severe, and we can integrate

$$\int_{\nu_0}^{\nu_\infty} \frac{d\nu}{\nu} \simeq -\int_{r_0}^{\infty} \frac{GM}{r^2c^2}\,dr,$$

where ν_0 and ν_∞ are the frequencies at r_0 and infinity, respectively. The result is

$$\ln\left(\frac{\nu_\infty}{\nu_0}\right) \simeq -\frac{GM}{r_0c^2},$$

which is valid when gravity is weak ($r_0/r_c = GM/r_0c^2 \ll 1$). This can be rewritten as

$$\frac{\nu_\infty}{\nu_0} \simeq e^{-GM/r_0c^2}. \tag{17.8}$$

Because the exponent is $\ll 1$, we use $e^{-x} \simeq 1 - x$ to get

$$\frac{\nu_\infty}{\nu_0} \simeq 1 - \frac{GM}{r_0c^2}. \tag{17.9}$$

This approximation shows the first-order correction to the frequency of the photon.

The exact result for the gravitational redshift, valid even for a strong gravitational field, is

$$\boxed{\frac{\nu_\infty}{\nu_0} = \left(1 - \frac{2GM}{r_0 c^2}\right)^{1/2}.} \tag{17.10}$$

When gravity is weak and the exponent in Eq. (17.8) is $\ll 1$, we use $(1-x)^{1/2} \simeq 1 - x/2$ to recover Eq. (17.9).

The gravitational redshift can be incorporated into the redshift parameter defined by Eq. (4.34), giving

$$z = \frac{\lambda_\infty - \lambda_0}{\lambda_0} = \frac{\nu_0}{\nu_\infty} - 1$$

$$= \left(1 - \frac{2GM}{r_0 c^2}\right)^{-1/2} - 1 \tag{17.11}$$

$$\simeq \frac{GM}{r_0 c^2}, \tag{17.12}$$

where Eq. (17.12) is valid only for a weak gravitational field.

To understand the origin of the gravitational redshift, imagine a clock that is constructed to tick once with each vibration of a monochromatic light wave. The time between ticks is then equal to the period of the oscillation of the wave, $\Delta t = 1/\nu$. Then according to Eq. (17.10), as seen from an infinite distance, the gravitational redshift implies that the clock at r_0 will be observed to run more slowly than an identical clock at $r = \infty$. If an amount of time Δt_0 passes at position r_0 outside a spherical mass, M, then the time Δt_∞ at $r = \infty$ is

$$\boxed{\frac{\Delta t_0}{\Delta t_\infty} = \frac{\nu_\infty}{\nu_0} = \left(1 - \frac{2GM}{r_0 c^2}\right)^{1/2}.} \tag{17.13}$$

For a weak field,

$$\frac{\Delta t_0}{\Delta t_\infty} \simeq 1 - \frac{GM}{r_0 c^2}. \tag{17.14}$$

We must conclude that *time passes more slowly as the surrounding spacetime becomes more curved*, an effect called **gravitational time dilation**. The gravitational redshift is therefore a consequence of time running at a slower rate near a massive object.

In other words, suppose two perfect, identical clocks are initially standing side by side, equally distant from a spherical mass. They are synchronized, and then one is slowly lowered below the other and then raised back to its original level. All observers will agree that when the clocks are again side by side, the clock that was lowered will be running behind the other because time in its vicinity passed more slowly while it was deeper in the mass's gravitational field.

Example 17.1.2. The white dwarf Sirius B has a radius of $R = 5.5 \times 10^6$ m and a mass of $M = 2.1 \times 10^{30}$ kg. The radius of curvature of the path of a horizontally traveling light beam near the surface of Sirius B is given by Eq. (17.5),

$$r_c = \frac{c^2}{g} = \frac{R^2c^2}{GM} = 1.9 \times 10^{10} \text{ m}.$$

The fact that $GM/Rc^2 = R/r_c \ll 1$ indicates that the curvature of spacetime is not severe. Even at the surface of a white dwarf, gravity is considered relatively weak in terms of its effect on the curvature of spacetime.

From Eq. (17.12), the gravitational redshift suffered by a photon emitted at the star's surface is

$$z \simeq \frac{GM}{Rc^2} = 2.8 \times 10^{-4}.$$

This is in excellent agreement with the measured gravitational redshift for Sirius B of $(3.0 \pm 0.5) \times 10^{-4}$.

To compare the rate at which time passes at the surface of Sirius B with the rate at a great distance, suppose that exactly one hour is measured by a distant clock. The time recorded by a clock at the surface of Sirius B would be *less* than one hour by an amount found using Eq. (17.14):

$$\Delta t_\infty - \Delta t_0 = \Delta t_\infty \left(1 - \frac{\Delta t_0}{\Delta t_\infty}\right) \simeq (3600 \text{ s}) \left(\frac{GM}{Rc^2}\right) = 1.0 \text{ s}.$$

The clock at the surface of Sirius B runs more slowly by about one second per hour compared to an identical clock far out in space.

The preceding experimental results (results obtained from tests of the equivalence principle) confirm the curvature of spacetime. In Section 17.2, we will learn that a freely falling particle takes the straightest possible path through curved spacetime.

17.2 ■ INTERVALS AND GEODESICS

We now consider the united concepts of space and time as expressed in *spacetime*, with four coordinates (x, y, z, t) specifying each *event*.[5] Einstein's crowning achievement was the deduction of his *field equations* for calculating the geometry of spacetime produced by a given distribution of mass and energy. His equations have the form

$$\boxed{\mathcal{G} = -\frac{8\pi G}{c^4}\mathcal{T}.} \tag{17.15}$$

[5]Nothing special (in fact, nothing at all) need happen at an event. Recall from Section 4.1 that an event is simply a location in spacetime identified by (x, y, z, t).

On the right is the *stress–energy tensor*, $\mathcal{T}$, which evaluates the effect of a given distribution of mass and energy on the curvature of spacetime, as described mathematically by the Einstein tensor, $\mathcal{G}$ (for $\mathcal{G}$ravity), on the left.[6] The appearance of Newton's gravitational constant, G, and the speed of light symbolizes the extension of relativity theory to include gravity. It is far beyond the scope of this book to delve further into this fascinating equation. We will be content merely to describe the curvature of spacetime around a spherical object of mass M and radius R, then demonstrate how an object moves through the curved spacetime it encounters.

Worldlines and Light Cones

Figure 17.12 shows three examples of some paths traced out in spacetime. In these *spacetime diagrams*, time is represented on the vertical axis, while space is depicted by the horizontal x–y plane. The third spatial dimension, z, cannot be shown, so this figure deals only with motion that occurs in a plane. The path followed by an object as it moves through spacetime is called its **worldline**. Our task will be to calculate the worldline of a freely falling object in response to the local curvature of spacetime. The spatial components of such a worldline describe the trajectory of a baseball arcing toward an outfielder, a planet orbiting the Sun, or a photon attempting to escape from a black hole.

The worldlines of photons in flat spacetime point the way to an understanding of the geometry of spacetime. Suppose a flashbulb is set off at the origin at time $t = 0$; call this event A. As shown in Fig. 17.13, the worldlines of photons traveling in the x–y plane form a **light cone** that represents a widening series of horizontal circular slices through the expanding spherical wavefront of light. The graph's axes are scaled so that the straight worldlines of light rays make 45° angles with the time axis.

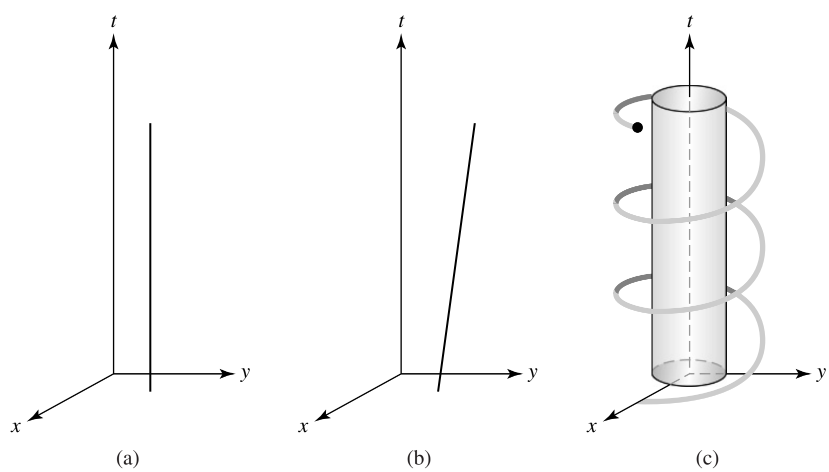


FIGURE 17.12 Worldlines for (a) a man at rest, (b) a woman running with constant velocity, and (c) a satellite orbiting Earth.

[6]Note that $E_{\text{rest}} = mc^2$ implies that both mass and energy contribute to the curvature of spacetime.

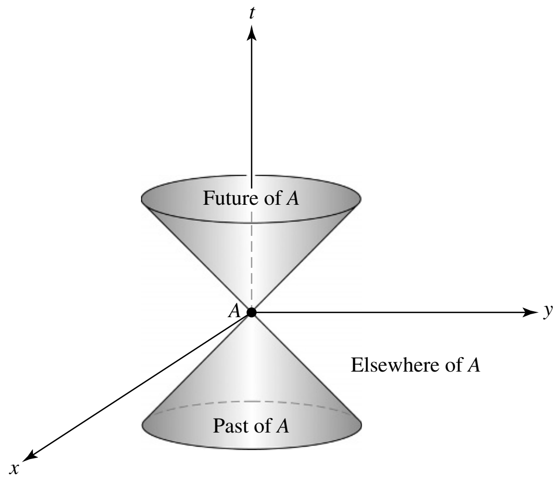


FIGURE 17.13 Light cones generated by horizontally traveling photons leaving the origin at time $t = 0$.

A massive object initially at event A must travel slower than light, so the angle between its worldline and the time axis must be less than 45°. Therefore the region inside the light cone represents the possible *future* of event A. It consists of all of the events that can possibly be reached by a traveler initially at event A—and therefore all of the events that the traveler could ever influence in a causal way.

Extending the diverging photon worldlines back through the origin generates a lower light cone. Within this lower light cone is the possible *past* of event A, the collection of all events from which a traveler could have arrived just as the bulb flashed. In other words, the possible past consists of the locations in space and time of every event that could possibly have caused the flashbulb to go off.

Outside the future and past light cones is an unknowable *elsewhere*, that part of spacetime of which a traveler at event A can have no knowledge and over which he or she can have no influence. It may come as a surprise to realize that vast regions of spacetime are hidden from us. You just can't get there from here.

In principle, every event in spacetime has a pair of light cones extending from it. The light cone divides spacetime into that event's future, past, and elsewhere. For any event in the past to have possibly influenced you, that event must lie within your past light cone, just as any event that you can ever possibly affect must lie within your future light cone. Your entire future worldline, your *destiny*, must therefore lie within your future light cone at every instant. Light cones act as spacetime horizons, separating the knowable from the unknowable.

Spacetime Intervals, Proper Time, and Proper Distance

Measuring the progress of an object as it moves along its worldline involves defining a "distance" for spacetime. Consider the familiar case of purely spatial distances. If two points have Cartesian coordinates

$$(x_1, y_1, z_1) \qquad \text{and} \qquad (x_2, y_2, z_2),$$

then the distance $\Delta\ell$ measured along the straight line between the two points in flat space is defined by

$$(\Delta\ell)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2.$$

The analogous measure of "distance" in spacetime is called the **spacetime interval** (or simply *interval* for short), first encountered in Problem 4.11. Let two events A and B have spacetime coordinates

$$(x_A, y_A, z_A, t_A) \qquad \text{and} \qquad (x_B, y_B, z_B, t_B),$$

measured by an observer in an inertial reference frame, S. Then the interval Δs measured along the straight worldline between the two events in flat spacetime is defined by

$$(\Delta s)^2 = [c(t_B - t_A)]^2 - (x_B - x_A)^2 - (y_B - y_A)^2 - (z_B - z_A)^2. \tag{17.16}$$

In words,

$$\begin{aligned}(\text{interval})^2 = {} & (\text{distance traveled by light in time } |t_B - t_A|)^2 \\ & - (\text{distance between events } A \text{ and } B)^2.\end{aligned}$$

This definition of the interval is very useful because, as shown in Problem 4.11, $(\Delta s)^2$ is *invariant* under a Lorentz transformation (Eqs. 4.16–4.19). An observer in another inertial reference frame, S', will measure the same value for the interval between events A and B; that is, $\Delta s = \Delta s'$.

Note that $(\Delta s)^2$ may be positive, negative, or zero. The sign tells us whether light has enough time to travel between the two events. If $(\Delta s)^2 > 0$, then the interval is *timelike* and light has more than enough time to travel between events A and B. An inertial reference frame S can therefore be chosen that moves along the straight worldline connecting events A and B so that the two events happen at the *same location* in S (at the origin, for example); see Fig. 17.14. Because the two events occur at the same place in S, the time measured between the two events is $\Delta s/c$. By definition, the time between two events that occur at the same location is the **proper time**, $\Delta\tau$, where

$$\boxed{\Delta\tau \equiv \frac{\Delta s}{c}} \tag{17.17}$$

(recall Section 4.3). The proper time is just the elapsed time recorded by a watch moving along the worldline from A to B. An observer in any inertial reference frame can use the interval to calculate the proper time between two events that are separated by a timelike interval.

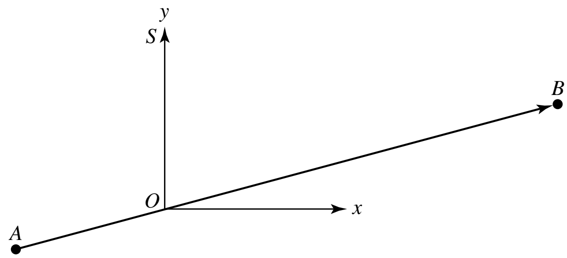


FIGURE 17.14 An inertial reference frame S moving along the timelike worldline connecting events A and B. Both events occur at the origin of S.

If $(\Delta s)^2 = 0$, then the interval is *lightlike* or *null*. In this case, light has exactly enough time to travel between events A and B. Only light can make the journey from one event to the other, and the proper time measured along a null interval is zero.

Finally, if $(\Delta s)^2 < 0$, then the interval is *spacelike*; light does not have enough time to travel between events A and B. No observer could travel between the two events because speeds greater than c would be required. The lack of absolute simultaneity in this situation, however, means that there are inertial reference frames in which the two events occur in the opposite temporal order, or even at the *same time*. By definition, the distance measured between two events A and B in a reference frame for which they occur simultaneously ($t_A = t_B$) is the **proper distance** separating them,[7]

$$\boxed{\Delta\mathcal{L} = \sqrt{-(\Delta s)^2}.} \tag{17.18}$$

If a straight rod were connected between the locations of the two events, this would be the *rest length* of the rod. An observer in any inertial reference frame can use this to calculate the proper distance between two events that are separated by a spacelike interval.[8]

The interval is clearly related to the light cones discussed in the foregoing paragraphs. Let event A be a flashbulb set off at the origin at time $t = 0$. The surfaces of the light cones, where the photons are at any time t, are the locations of all events B that are connected to A by a null interval. The events within the future and past light cones are connected to A by a timelike interval, and the events that occur elsewhere are connected to A by a spacelike interval.

The Metric for Flat Spacetime

Returning to three-dimensional space for a moment, it is obvious that a path connecting two points in space doesn't have to be straight. Two points can be connected by infinitely many curved lines. To measure the distance along a curved path, $\mathcal{P}$, from one point to the

[7]In Section 4.3, when the emphasis was on length rather than distance, this was called the *proper length*. The terms may be used interchangeably, depending on the context.

[8]For both proper time and proper distance, the term *proper* has the connotation of "measured by an observer who is right there, moving along with the clock or the rod."

other, we use a differential distance formula called a **metric**,

$$(d\ell)^2 = (dx)^2 + (dy)^2 + (dz)^2.$$

Then $d\ell$ may be integrated along the path $\mathcal{P}$ (a *line integral*) to calculate the total distance between the two points,

$$\Delta\ell = \int_1^2 \sqrt{(d\ell)^2} = \int_1^2 \sqrt{(dx)^2 + (dy)^2 + (dz)^2} \quad (\text{along } \mathcal{P}).$$

The distance between two points thus depends on the path connecting them. Of course, the *shortest* distance between two points in flat space is measured along a straight line. In fact, we can *define* the "straightest possible line" between two points as the path for which $\Delta\ell$ is a *minimum*.

Similarly, a worldline between two events in spacetime is not required to be straight; the two events can be connected by infinitely many curved worldlines. To measure the interval along a curved worldline, $\mathcal{W}$, connecting two events in spacetime with no mass present, we use the **metric for flat spacetime**,

$$\boxed{(ds)^2 = (c\,dt)^2 - (d\ell)^2 = (c\,dt)^2 - (dx)^2 - (dy)^2 - (dz)^2.} \tag{17.19}$$

Then ds is integrated to determine the total interval along the worldline $\mathcal{W}$,

$$\Delta s = \int_A^B \sqrt{(ds)^2} = \int_A^B \sqrt{(c\,dt)^2 - (dx)^2 - (dy)^2 - (dz)^2} \quad (\text{along } \mathcal{W}).$$

The interval is still related to the proper time measured along the worldline by Eq. (17.17). *The interval measured along any timelike worldline divided by the speed of light is always the proper time measured by a watch moving along that worldline.* The proper time is zero along a null worldline and is undefined for a spacelike worldline.

In flat spacetime, the interval measured along a straight timelike worldline between two events is a *maximum*. Any other worldline between the same two events will not be straight and will have a smaller interval. For a massless particle such as a photon, all worldlines have a null interval (so $\int \sqrt{(ds)^2} = 0$).

The maximal character of the interval of a straight worldline in flat spacetime is easily demonstrated. Figure 17.15 is a spacetime diagram showing two events, A and B, that occur at times t_A and t_B. The events are observed from an inertial reference frame, S, that moves from A to B, chosen such that the two events occur at the origin of S. The interval measured along the straight worldline connecting A and B is

$$\begin{aligned}
\Delta s(A \to B) &= \int_A^B \sqrt{(ds)^2} \\
&= \int_A^B \sqrt{(c\,dt)^2 - (dx)^2 - (dy)^2 - (dz)^2} \\
&= \int_{t_A}^{t_B} c\,dt = c(t_B - t_A).
\end{aligned}$$

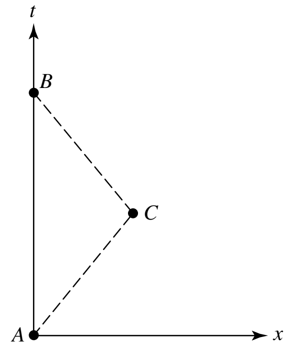


FIGURE 17.15 Worldlines connecting events A and B.

Now consider the interval measured along another worldline connecting A and B that includes event C, which occurs at $(x, y, z, t) = (x_C, 0, 0, t_C)$. In this case,

$$\begin{aligned}\Delta s(A \to C \to B) &= \int_A^C \sqrt{(ds)^2} + \int_C^B \sqrt{(ds)^2} \\ &= \int_A^C \sqrt{(c\,dt)^2 - (dx)^2 - (dy)^2 - (dz)^2} \\ &\quad + \int_C^B \sqrt{(c\,dt)^2 - (dx)^2 - (dy)^2 - (dz)^2}.\end{aligned}$$

Using $dx/dt = v_{AC}$ for the constant velocity along worldline $A \to C$ in the first integral, and $dx/dt = v_{CB}$ for the constant velocity along $C \to B$ in the second integral, leads to

$$\begin{aligned}\Delta s(A \to C \to B) &= (t_C - t_A)\sqrt{c^2 - v_{AC}^2} + (t_B - t_C)\sqrt{c^2 - v_{CB}^2} \\ &< c(t_C - t_A) + c(t_B - t_C) \\ &< \Delta s(A \to B).\end{aligned}$$

Thus the straight worldline has the longer interval. Any worldline connecting event A and B can be represented as a series of short segments, so we can conclude that the interval Δs is indeed a maximum for the straight worldline.

Curved Spacetime and the Schwarzschild Metric

In a spacetime that is curved by the presence of mass, the situation is slightly more complicated. Even the "straightest possible worldline" will be curved. These straightest possible worldlines are called **geodesics**. In flat spacetime a geodesic is a straight worldline.

In curved spacetime, a timelike geodesic between two events has either a *maximum* or a *minimum* interval. In other words, the value of Δs along a timelike geodesic is an *extremum*, either a maximum or a minimum, when compared with the intervals of nearby worldlines between the same two events.[9] In the situations we will encounter in this chapter, the intervals of timelike geodesics will be maxima. A massless particle such as a photon follows a *null geodesic*, with $\int \sqrt{(ds)^2} = 0$.[10] Einstein's key realization was that *the paths followed by freely falling objects through spacetime are geodesics*.

We are now prepared to deal with the effect of mass on the geometry of spacetime, based on the three fundamental features of general relativity:

- Mass acts on spacetime, telling it how to curve.
- Spacetime in turn acts on mass, telling it how to move.
- Any freely falling particle (including a photon) follows the straightest possible worldline, a geodesic, through spacetime. For a massive particle, the geodesic has a *maximum* or a *minimum* interval, while for light, the geodesic has a *null* interval.

These components of the theory will allow us to describe the curvature of spacetime around a massive spherical object and to determine how another object will move in response, whether it is a satellite orbiting Earth or a photon orbiting a black hole. For situations with spherical symmetry, it will be more convenient to use the familiar spherical coordinates (r, θ, ϕ) instead of Cartesian coordinates. The metric between two nearby points in flat space is then

$$(d\ell)^2 = (dr)^2 + (r\,d\theta)^2 + (r \sin\theta\, d\phi)^2, \tag{17.20}$$

and the corresponding expression for the flat spacetime metric is

$$\boxed{(ds)^2 = (c\,dt)^2 - (dr)^2 - (r\,d\theta)^2 - (r \sin\theta\, d\phi)^2.} \tag{17.21}$$

Of course, spacetime will not be flat in the vicinity of a massive object. The specific situation to be investigated here is the motion of a particle through the curved spacetime produced by a massive sphere. It could be a planet, a star, or a black hole. The first task is to calculate how this massive object acts on spacetime, telling it how to curve. This requires a description of the metric for this curved spacetime that will replace Eq. (17.21) for a flat spacetime.

Before presenting this metric, we must emphasize that the variables r, θ, ϕ, and t that appear in the expression for the metric are the *coordinates* used by an observer at rest a great ($\simeq$ infinite) distance from the origin. In the absence of a central mass at the origin, r would be the distance from the origin, and differences in r would measure the distance between points on a radial line. The time t measured by clocks scattered throughout the coordinate system would remain synchronized, advancing everywhere at the same rate.

[9] In fact, a calculation of the intervals of nearby worldlines would show that the interval of a timelike geodesic corresponds to a maximum, a minimum, or an inflection point. You are referred to Section 13.4 of Misner, Thorne, and Wheeler (1973) for an interesting discussion of geodesics as worldlines of extremal proper time.

[10] The extremal principle for intervals cannot be directly applied to find the straightest possible worldline for a photon, since its interval is always null. However, the straightest possible worldline for a massless particle is the same as that for a massive particle in the limit of a vanishingly small mass as its velocity $v \to c$.

Now we place a sphere of mass M and radius R (which will be called a "planet") at the origin of our coordinate system. Some care must be taken in laying out the radial coordinate. The origin (which is inside the sphere) should not be used as a point of reference, and so we will avoid defining r as "the distance from the origin." Instead, imagine a series of nested concentric spheres centered at the origin. The surface area of a sphere can be measured without approaching the origin, so the coordinate r will be defined by the surface of that sphere having an area $4\pi r^2$. With this careful approach, we will find that these coordinates can be used with the metric for curved spacetime to measure distances in space and the passage of time near this massive sphere. As an object moves through this curved spacetime, its **coordinate speed** is just the rate at which its spatial coordinates change.

At a large distance ($r \simeq \infty$) from the planet, spacetime is essentially flat, and the gravitational time dilation of a photon received from the planet is given by Eq. (17.13). From this, it might be expected that $\sqrt{1 - 2GM/rc^2}$ would play a role in the metric for the spacetime surrounding the planet. Furthermore, recall from Section 17.1 that the stretching of space and the slowing down of time contribute equally to delaying a light beam's passage through curved spacetime. This provides a hint that the same factor will be involved in the metric's radial term. The angular terms are the same as those in Eq. (17.21) for flat spacetime.

These effects are indeed present in the metric that describes the curved spacetime surrounding a spherical mass, M. In 1916, just two months after Einstein published his general theory of relativity, the German astronomer Karl Schwarzschild (1873–1916) solved Einstein's field equations to obtain what is now called the **Schwarzschild metric**:

$$\boxed{\begin{aligned}(ds)^2 = {} & \left(c\,dt\sqrt{1 - 2GM/rc^2}\right)^2 - \left(\frac{dr}{\sqrt{1 - 2GM/rc^2}}\right)^2 \\ & - (r\,d\theta)^2 - (r\sin\theta\,d\phi)^2.\end{aligned}} \tag{17.22}$$

There is no other, easier way to obtain the Schwarzschild metric, so we must be content with the foregoing heuristic description of its terms.

It is important to realize that the Schwarzschild metric is the spherically symmetric *vacuum solution* of Einstein's field equations. That is, it is valid only in the empty space *outside* the object. The mathematical form of the metric is different in the object's interior, which is occupied by matter.

The Schwarzschild metric contains all of the effects considered in the last section. The "curvature of space" resides in the radial term. The radial distance measured simultaneously ($dt = 0$) between two nearby points on the same radial line ($d\theta = d\phi = 0$) is just the proper distance, Eq. (17.18),

$$d\mathcal{L} = \sqrt{-(ds)^2} = \frac{dr}{\sqrt{1 - 2GM/rc^2}}. \tag{17.23}$$

Thus the spatial distance $d\mathcal{L}$ between two points on the same radial line is *greater* than the coordinate difference dr. This is precisely what is represented by the stretched grid lines in the rubber sheet analogy of the previous section. The factor of $1\big/\sqrt{1 - 2GM/rc^2}$ must be included in any calculation of spatial distances. This is analogous to using a topographic